

STATEMENT LETTER

Monash ID: 33462135

Date Issued: 10 October 2025

Student Name: Reza Darmawan

Issue No.: 90/SL-SAS/MU_I/X/2025

Monash University, Indonesia**START DATE: 02 October 2023**

Course name	Master of Data Science
Registered Provider	Monash University Indonesia
Campus	Bumi Serpong Damai, South Tangerang
Monash course code	C6009
Credits points required to complete this course	96 credit points are required to complete this course (No credits granted)
Credit points completed	96 credit points
Grade Point Average (GPA) and Weighted Average Mark (WAM) upon course completion	Grade Point Average (GPA): 3.625 Weighted Average Mark (WAM): 83.188

Monash University in Indonesia confirms that the student mentioned above has completed all the required units to obtain a degree in Master of Data Science. The student was course completed by 17 October 2025. Monash University students are awarded official certificates (testamur) during the Graduation Ceremony, where the degree is formally conferred and officially awarded.

This letter serves as temporary confirmation on their course completion and can be used to support employment or further studies until the official degree conferral is due.

All the information provided is correct to the best of our knowledge. Should further information or clarification about this letter be needed, please reach out to ask.monashindonesia@monash.edu.



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Annex 1. Graduation Process Information

COURSE COMPLETION:

After results are released for the final teaching period, faculty will need to confirm that students have completed all the requirements (including non-credit units and activities) of their course, finalise the students' mark, and mark students as officially course completed. Once students are course completed, students can:

- process their application to graduate
- students' transcripts will show a complete record of their marks and grades
- students can order a complete academic transcript
- students can order a proof of qualification letter (currently available only to Monash Australia students. This course completion statement letter serves as an interim replacement for a proof of qualification letter for Monash Indonesia students).

GRADUATION CEREMONIES

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Eligibility

All students who are nearing course completion are eligible to apply to graduate. Students who hold this course completion letter are eligible to apply.



ACADEMIC RECORD

Student Name: Reza Darmawan

Student ID: 33462135

Date of Issue: 20-OCT-2025

QUALIFIED ON 17 OCTOBER 2025 FOR MASTER OF DATA SCIENCE

MASTER OF DATA SCIENCE

		Teaching Period	Credit Points	Mark	Grade
2025					
ITI5126	MASTERS THESIS PART 1	MI-T2	-	-	NE
ITI5201	MACHINE LEARNING	MI-T2	6	91	HD
ITI5122	PROFESSIONAL PRACTICE	MI-T3	6	76	D
ITI5127	MASTERS THESIS PART 2	MI-T3	-	-	NE
ITI5128	MASTERS THESIS FINAL	MI-T4	18	81	HD
ITI5202	DATA PROCESSING FOR BIG DATA	MI-T4	6	67	C
2024					
ITI9004	MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE	MI-T2	6	87	HD
ITI9137	INTRODUCTION TO COMPUTER ARCHITECTURE AND NETWORKS	MI-T2	6	94	HD
ITI5125	IT RESEARCH METHODS	MI-T3	6	87	HD
ITI5147	DATA EXPLORATION AND VISUALISATION	MI-T3	6	80	HD
ITI5145	INTRODUCTION TO DATA SCIENCE	MI-T4	6	92	HD
ITI5197	STATISTICAL DATA MODELLING	MI-T4	6	64	C
ITI5057	PROJECT MANAGEMENT	MI-T1	6	75	D
ITI5196	DATA WRANGLING	MI-T1	6	94	HD
2023					
ITI9132	INTRODUCTION TO DATABASES	MI-T1	6	89	HD
ITI9136	ALGORITHMS AND PROGRAMMING FOUNDATIONS IN PYTHON	MI-T1	6	92	HD

Course	GPA	WAM
MASTER OF DATA SCIENCE	3.625	83.188

For an explanation of how the Grade Point Average (GPA) and Weighted Average Mark (WAM) are calculated, please refer to <http://adm.monash.edu.au/service-centre/academic-transcripts.html>



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KEY TO RESULTS

From 2010

HD	High Distinction	80-100	DEF	Deferred Assessment
D	Distinction	70-79	E	Exempt
C	Credit	60-69	FP	Faculty Pass
P	Pass	50-59	NA	Not Applicable
N	Fail	0-49	NAS	Non-Assessed
NH	Hurdle Fail	45	NE	Not Examinable
			NGO	Fail
HI/H1	First Class Honours	80+	NS	Supplementary Assessment Granted
IIIA/H2A	Second Class Honours Division A	70-79	NSR	Not Satisfied Requirements
IIIB/H2B	Second Class Honours Division B	60-69	M	Merit
III	Third Class Honours	50-59	PGO	Pass Grade Only (no higher grade available)
			SFR	Satisfied Faculty Requirements
			WDN	Withdrawn
			WH	Withheld
			WI	Withdrawn Incomplete
			WN	Withdrawn Fail
			WNGO	Withdrawn Fail

In 2020, Monash University introduced a modified grading scale system in response to the COVID-19 pandemic. More information on grades awarded during this period, including how these modifications relate to a student's Grade Point Average and Weighted Average Mark, can be found at monash.edu/exams/results-legend.html

Pre-2010

Details of pre-2010 results can be found at monash.edu/exams/results-legend.html

This is a valuable document and should be kept in a safe place.

**IndoMPF: Dataset and Tabular-Graph Numerical Reasoning
Baseline Model for Generating Notes to Government
Financial Statements**



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Part 1

General Literature Review

Generating Notes to Government Financial Statements using Multi-step Reasoning and Tabular Graph-Enhanced Language Model

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Prepared for
Master's Thesis Part 1
(IT15126)



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1. Introduction

Notes to financial statements accompanies a financial statement to enhance its understandability and to avoid misunderstanding. Notes to financial statements from private entities commonly consist of general Information about reporting entity, accounting policy, and explanation of each financial statement post. Government financial statements, on the other hand, embody government responsibility to public financial management. Therefore, particularly in Indonesia, as required in the Government Accounting Standard issued by KSAP, there are two extra components to the notes, i.e. budget information, summary of fiscal/monetary policy and macroeconomic situation.

The process of manual drafting notes to financial statements is time intensive. It is also prone to content inaccuracy and incompleteness of information disclosure, which requires additional control process e.g. financial statement review by government internal auditor unit.

The current development of Language Model that allows for various text-generation tasks brings the potential of automating this business process. Many of the data sources are in tabular format, therefore table-to-text generation downstream task may be suitable.

However, language models often struggle with hallucination problems. For example, Chapman et al (2022) has attempted to generate notes to financial statements by proposing the utilization of BERT2BERT Transformer architecture for table-to-text generation task, using cell-highlighting approach. This solution shows promising results, but with BLEU score of 63.3 hallucination problems still loom. Language model hallucination is a problem when implementing language model for tasks related to financial reporting, particularly for government financial statements which constitute as the symbol of public financial management accountability.

B.2.1.2.1. Natural Resources Revenue
 Realization of Natural Resources Revenue in FY 2023 amounted to Rp254,203,031,073,614 or reached 113.83 percent of the amount set in the State Budget of Rp223,312,141,813,000. Natural Resources revenue in FY 2023 was Rp14,567,865,197,170 smaller or 5.42 percent lower compared to the realization in FY 2022 which amounted to Rp268,770,896,270,784 which was mainly due to a decrease in oil and natural gas revenue.

Account Code	Account Description	Budget	Budget Realization 2023	% of Budget	Budget Realization 2022	Increase/ Decrease	% Increase/ Decrease
421	REVENUE FROM NATURAL RESOURCES	223,312,141,813,000	254,203,031,073,614	113.83%	268,770,896,270,784	(14,567,865,197,170)	-5.42%
4211	REVENUE FROM OIL	76,351,120,000,000	87,408,085,465,410	114.48%	111,992,940,458,410	(24,584,854,992,994)	-21.95%
4212	REVENUE FROM NATURAL GAS	27,250,340,000,000	26,769,670,492,312	105.58%	36,705,949,688,708	(7,936,279,196,396)	-21.62%
4213	REVENUE FROM MINERAL AND COAL MINING	109,974,491,731,000	129,126,791,868,349	117.42%	110,786,795,000,708	18,339,996,867,641	16.55%
4214	REVENUE FROM FORESTRY	5,681,650,000,000	5,429,603,320,398	95.56%	5,804,080,517,216	(324,477,196,818)	-6.45%
4215	REVENUE FROM FISHERY	1,628,000,000,000	636,893,353,269	39.12%	1,192,945,829,371	(556,052,476,102)	-46.61%
4216	REVENUE FROM GEOTHERMAL	2,426,540,000,000	2,831,186,573,870	116.68%	2,288,184,776,371	543,001,797,499	23.73%

Task A: Summarizing a row in the table
Task B: Explanation based on the value from other rows

Figure 1 Explaining the cause of increase or decrease in financial statement

A more complex task in writing notes to financial statements, beyond accurate summarization and explanation of certain facts from a table, is when the explanation requires numerical and comparative reasoning. For example, **Figure 1** depicts an example from Indonesia Central Government Financial Statement for the fiscal year 2023, of how certain explanation in notes to financial statement requires additional analysis beyond fact summarization. As can be observed, the decrease in revenue from natural resources requires a further breakdown regarding what other

balances contribute to this decrease in revenue. This is complex reasoning when, if it is done by humans, there are several steps of identification that need to be done. First is to recognize that revenue from natural resources consists of several subtypes of revenue e.g. oil, natural gas, mining, and geothermal. Then, among these subtypes of revenue, identify which subtypes of revenue with the highest year on year decrease, and frame it as the main contributor to the decrease in revenue from natural resources.

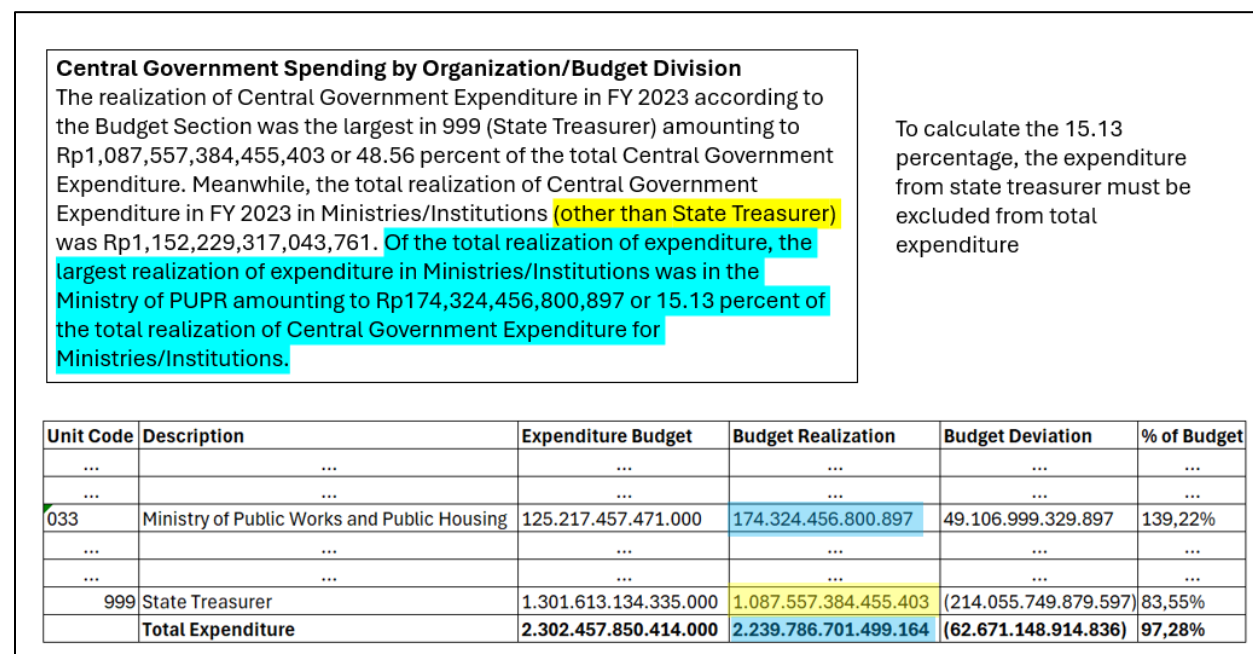


Figure 2 Calculating the percentage of expenditure, after excluding certain amount

Another example of complex reasoning in government notes to financial statements is calculating the percentage of expenditure by the Ministry with the highest amount as illustrated in **Figure 2**. However, what makes this complicated is that a certain expenditure needs to be first excluded, prior to calculating the percentage, i.e. expenditure from State Treasurer. Retrieving the correct cell and identifying step by step numerical reasoning is a challenge for language model, particularly for encoder-decoder language models e.g. T5 (Raffel et al., 2020) and BERT (Devlin et al., 2019) that are pretrained for text-to-text generation.

Numerical reasoning remains a challenge in table-to-text generation task, in which the model proposed struggle with generating texts beyond simple retrieval of superficial facts from the table (Alonso et al., 2024; Chapman et al., 2022; Guo et al., 2024; Parikh et al., 2020). Several approaches have endeavored to introduce numerical reasoning in table-to-text generation, which some works are limited to single-step reasoning (Nararatwong et al., 2022a; Suadaa et al., 2021; Zhu et al., 2021) and other have advanced to handling multi-step reasoning (Chen et al., 2021; C. Li et al., 2022).

In the other hand, since most language models are pretrained for text-to-text generation, table linearization still becomes the mainstream way of representing table to the language model (Chapman et al., 2022; C. Li et al., 2022; Murakami et al., 2017; Parikh et al., 2020; Puduppully et al., 2022; Puduppully & Lapata, 2021; Suadaa et al., 2021; Zhu et al., 2021). However, Li et al (2024) argue that linear representation is an oversimplification of data structure, and therefore several works

propose alternative representation of table structure e.g. graph (S. Li et al., 2024; Nararatwong et al., 2022a; Wang et al., 2022) and image (Alonso et al., 2024).

Therefore, a literature review of past works related to table-to-text generation needs to be conducted, particularly covering the following themes:

1. **Approaches to table-to-text generation.** Identifying the various approaches from previous works for table-to-text generation tasks and how each approach handles numerical reasoning from the table.
2. **Table reasoning dataset.** Identify the publicly available datasets from previous works to enhance the models for table-to-text tasks. Also identify whether the available datasets are sufficient to imbue reasoning ability in the models.

2. Literature Review

2.1. Table-to-text Generation Approaches

In general, table-to-text generation refers to the downstream task that aims to generate textual outputs based on structured, non-linguistic input data e.g. database, spreadsheet, and table image. In the following part, I review a variety of approaches from previous work in addressing table-to-text generation tasks. These approaches vary in terms of how the guidance is provided to the model in generating text and the components needed in the dataset.

2.1.1. Highlighted cells

This approach works by highlighting certain cells that may provide guidance to the model in generating text based on the table. Despite cells highlighting appears as a visual aspect of a table, most previous works in this approach require table linearization as the input to the model (Chapman et al., 2022; Parikh et al., 2020).

Table Title: Gabriele Becker					
Section Title: International Competitions					
Table Description: None					
Year	Competition	Venue	Position	Event	Notes
Representing Germany					
1992	World Junior Championships	Seoul, South Korea	10th (semis)	100 m	11.83
1993	European Junior Championships	San Sebastián, Spain	7th	100 m	11.74
			3rd	4x100 m relay	44.60
1994	World Junior Championships	Lisbon, Portugal	12th (semis)	100 m	11.66 (wind: +1.3 m/s)
			2nd	4x100 m relay	44.78
1995	World Championships	Gothenburg, Sweden	7th (q-finals)	100 m	11.54
			3rd	4x100 m relay	43.01
Original Text: After winning the German under-23 100 m title, she was selected to run at the 1995 World Championships in Athletics both individually and in the relay.					
Text after Deletion: she at the 1995 World Championships in both individually and in the relay.					
Text After Decontextualization: Gabriele Becker competed at the 1995 World Championships in both individually and in the relay.					
Final Text: Gabriele Becker competed at the 1995 World Championships both individually and in the relay.					

Figure 3 Example of highlighted table in ToTTo paper (Parikh et al., 2020)

ToTTo (Parikh et al., 2020) pioneered in the table-to-text generation task that utilized highlighted cells on input table (as illustrated by **Figure 3**) as a guidance for Transformer-based Bert2Bert model. Using Wikipedia as the main data source, annotator-highlighted tables are used as input and annotator-

modified texts are used as target, both constitutes as ToTTo dataset. This paper addresses an important finding that the usage of only certain parts of the table (subtable) leads to better performance compared to using the entire table as the source, even though this limits model's ability to reason in the context of table structure. Another important finding in ToTTo paper (Parikh et al., 2020) is that adding table metadata, i.e. Wikipedia page title and section title, to the source table improve model performance.

As an implementation of ToTTo paper (Parikh et al., 2020) in financial reporting domain, Chapman et al (2022) applies table-to-text generation to create financial statements from tabular data using Transformer-based Bert2Bert model. Since the authors are unable to get their hands on the financial statement data source, they only use the detailed tables from the private companies' notes to financial statements and aim to recreate the written section explaining the tables. These tables are linearized, in which each cell is separated using special tokens representing table component e.g. row/column header, section title, row/column content, and marking for highlighted cells.

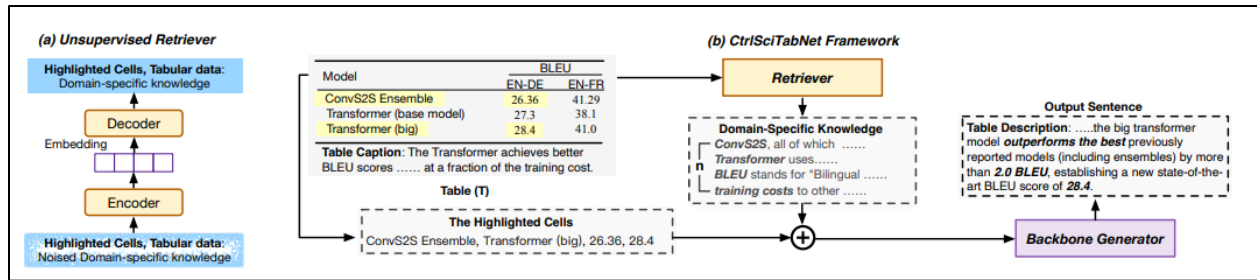


Figure 4 Retriever-Generator approach in CTRLSciTabNet (Guo et al., 2024)

CTRLSciTabNet (Guo et al., 2024) aims to generate text based on highlighted table, using retriever-generator framework. The retriever part is obtaining relevant information based on the input table from the previously constructed knowledge base. And based on the input table and additional context retrieved from knowledge base, coherent sentence and sensible description is produced by the model. This retriever-generator framework is illustrated in **Figure 4**.

However, hallucination cases are still found within the text generated from this approach, which from the provided examples in previous works (Chapman et al., 2022; Parikh et al., 2020) includes numerical reasoning and inaccurate retrieval. Chapman et al. (2022) argue that this hallucination is caused by overlapping tables, i.e. several samples that refer to the same table. In the other hand, Parikh et al (2020) pointed out that hallucinations can be caused by diversity of table structures, a particular characteristic of ToTTo dataset used in the paper.

In the other hand, this approach relies on manual highlight or annotation to mark relevant cells in the table for more accurate text generation (Chapman et al., 2022), This requirement may hinder optimal value addition to practical implementation of this solution.

2.1.2. Content Planning

A set of notable works attempt to approach table-to-text generation task by proposing the idea of content planning. This approach guides model output by defining required information from the table based on the reference text (Murakami et al., 2017; Puduppully et al., 2022; Puduppully & Lapata, 2021; Suadaa et al., 2021).

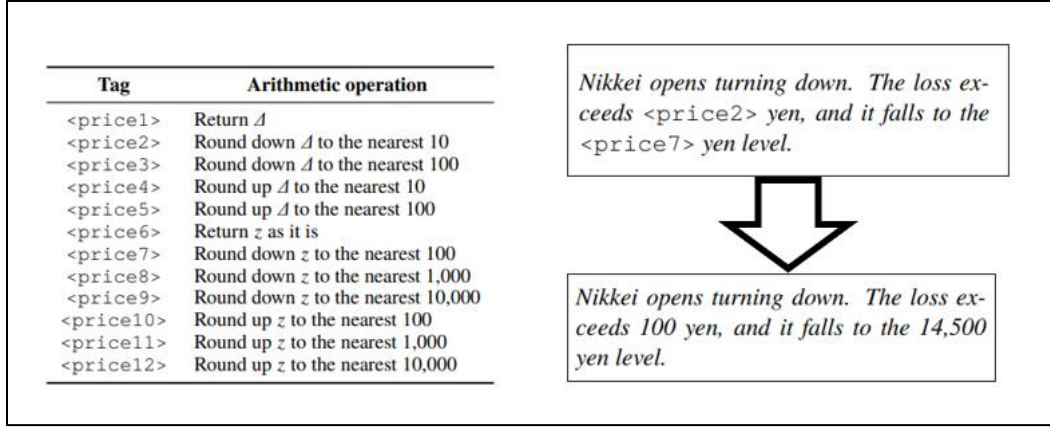


Figure 5 Murakami et al (2017) generate text from table using generalization tags representing arithmetic operations

Murakami et al (2017), in order to generate market comments based on provided series of stock prices data, applies generalization tags that represent arithmetic operations to stock price, e.g. round down to nearest hundred and round up to nearest thousand. These generalization tags are added as vocabularies to be used in the text generation model. In the output text, these generalization tags will be replaced with the result of numerical operation it represents. This approach is illustrated in **Figure 5**.

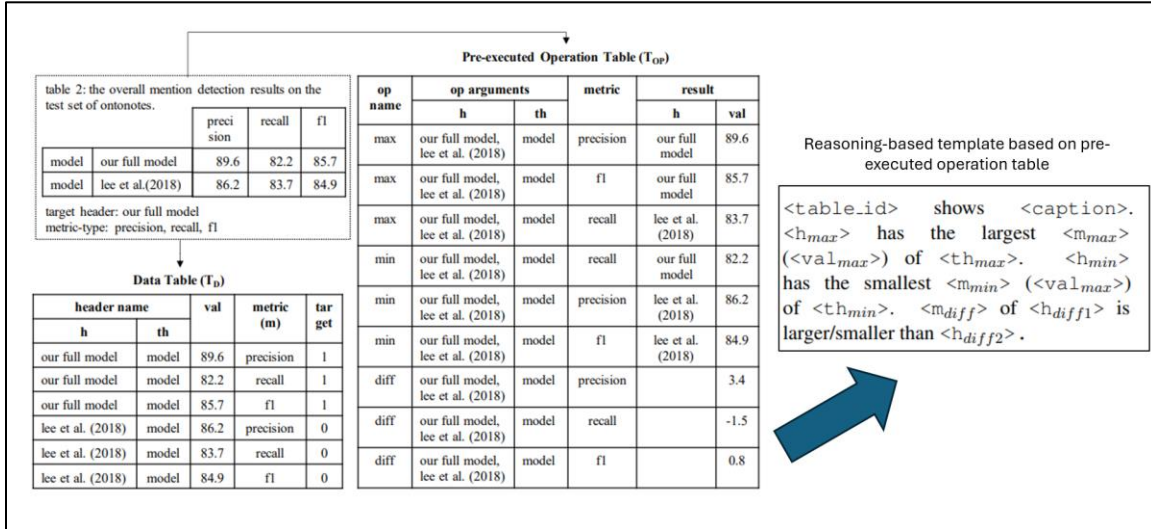


Figure 6 Suadaa et al (2021) uses pre-executed operation table containing numerical operation, and create reasoning based template as input to language model

Another idea is to utilize a template-guided representation for table. Suadaa et al (2021) use transfer learning of pre-trained Transformer, i.e. GPT2 (Radford et al., 2019) and T5 (Raffel et al., 2020), by providing template-based input representation that consists of two parts. The first part is a natural language sentence of the table, representing the semantics of data. And the second part is the natural language sentence of pre-executed operation table containing basic numerical operation based on the table, i.e. maximum, minimum, and difference between values. Template-based representation of table is utilized to solve representation bias because of the difference in

information representation compared to what the model has seen during pre-training phase (Kale & Rastogi, 2020). This approach is illustrated in **Figure 6**.

V(Orioles), V(Royals), V(C.Mullins), V(J.Villar), V(W.Merrifield), V(R.O'Hearn), V(A.Cashner), V(B.Keller), V(H.Dozier), ..., V(1-T), V(1-B), V(2-T), V(2-B), V(3-T), V(3-B), ...	V(Royals) V(Orioles), V(Orioles) V(C.Mullins), V(Orioles) V(J.Villar), V(Royals) V(W.Merrifield), V(Royals) V(R.O'Hearn), V(Orioles) V(A.Cashner), V(Royals) V(B.Keller), ..., V(C.Mullins) V(Royals) V(Orioles), V(J.Villar) V(Royals) V(Orioles), ...
V(B.Keller)<P>V(B.Keller) V(C.Mullins) V(Royals) V(Orioles)<P>V(B.Keller)<P> V(R.O'Hearn) V(W.Merrifield) V(H.Dozier) V(C.Gallagher) <P>V(4-B, 5-B) <P> V(6-T) <P>	

Figure 7 (Puduppully et al., 2022; Puduppully & Lapata, 2021) represents table content as verbalization of entities (top), then paragraph templates are constructed based on these entity verbalizations (bottom).

Other works (Puduppully et al., 2022; Puduppully & Lapata, 2021) present a different approach to content planning, which includes two stages of work where salient entities and events from the structured data are first verbalized. This verbalization of entities basically contains the linearized table. Then, a macro plan of paragraph which represents the paragraph sentences being mapped to the verbalized entities is generated. This approach is illustrated in **Figure 7**. However, the limitation for this method is numerical operation and sentence coherence produced.

2.1.3. Program Generator

There have been several approaches to create a model that can generate a suitable program to obtain information from a table that requires numerical reasoning. Among the previous works, several implementations require table linearization or flattening (Z. Chen et al., 2021; C. Li et al., 2022; Zhu et al., 2021). Other work use tabular graph representation as input (Nararatwong et al., 2022a).

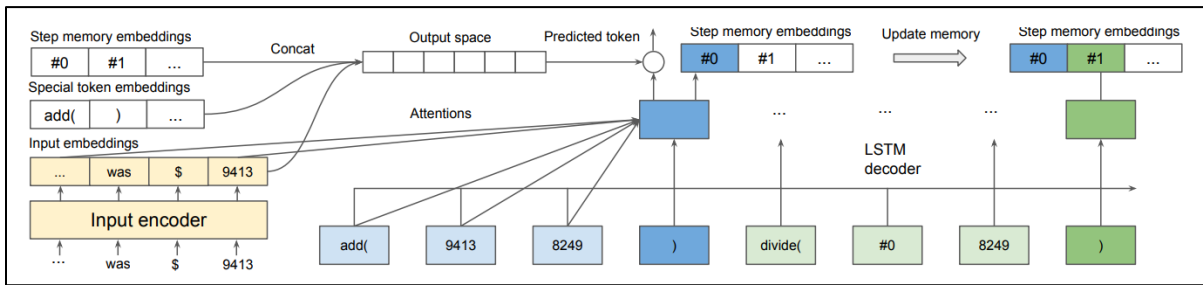


Figure 8 FinQANet (Chen et al., 2021) architecture for multi-step numerical reasoning program generator

FinQANet (Chen et al., 2021) linearizes the table input then implements an approach that enables multi-step numerical reasoning. As illustrated in **Figure 8**, using LSTM for decoding, in each decoding step the following token embedding are used as the input i.e. 1.) input table and question token, 2.) special tokens that consist of function names, and 3.) step memory tokens storing the result of calculation from the previous steps. The output to each decoding step is combined with attention vector of the input and attention vector of the decoding history, and form context vector. And this context vector is updated in each operation step.

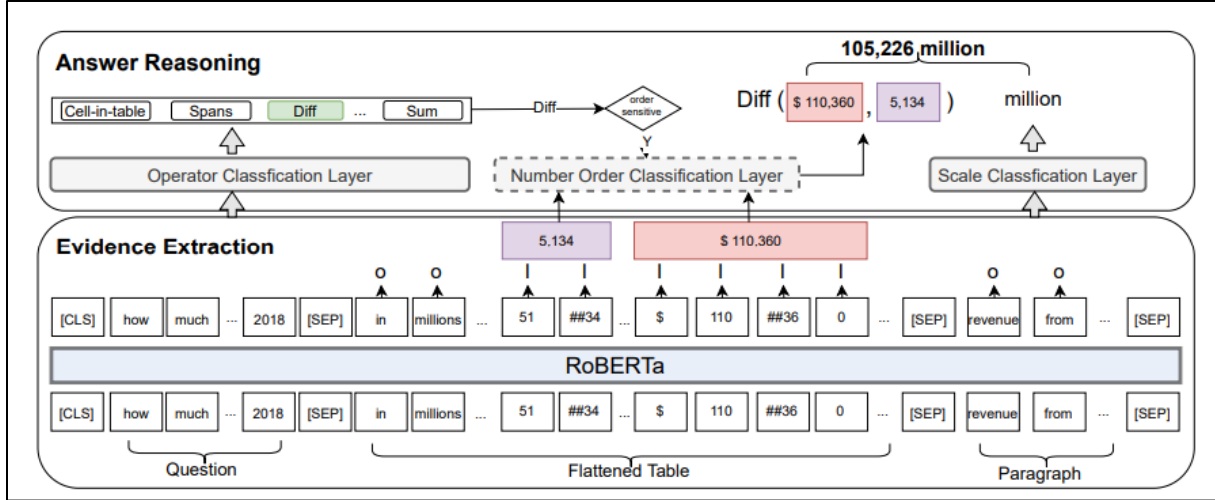


Figure 9 TagOp (Zhu et al., 2021), a multi task learning approach that consists of several classification layers

TagOp (Zhu et al., 2021) has different approach for program generator by offering multi-task learning (as illustrated in **Figure 9**), which solution consist of four classification layers i.e. evidence extraction classifier, operator classifier, number order classifier, and number scaling classifier. These classification layers do separate tasks required for numerical reasoning before the model can arrive in the final answer based on the query.

The pioneering work by TagOp (Zhu et al., 2021) leads to other enhancements, as reflected in subsequent works. GANO (Nararatwong et al., 2022a) for instance, uses graph neural network (GNN) as evidence extraction classifier while also representing the input table as tabular graph. The tabular graph will be explained in more detail in section 2.1.4. KIQA (Nararatwong et al., 2022b) improved TagOp architecture by adding knowledge injection layer, aiming to improve the model’s symbolic knowledge inherently obtained during pre-training, by feeding additional entity information to the language model.

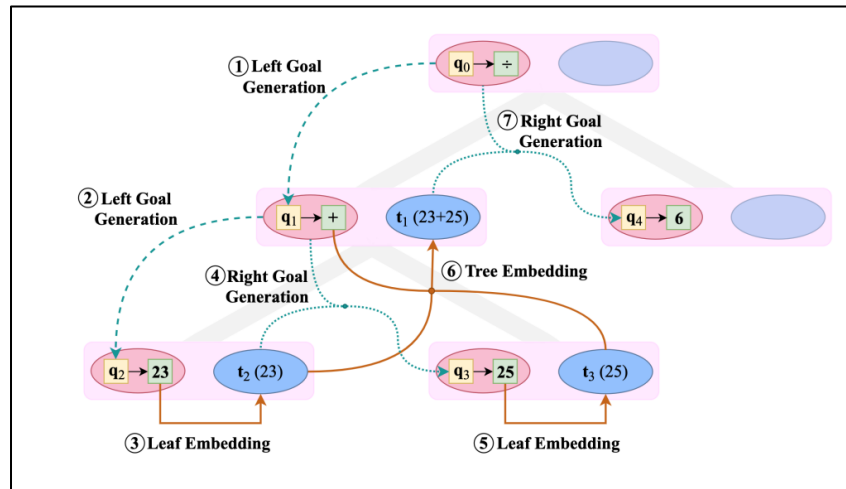


Figure 10 Tree structured neural model by (Xie & Sun, 2019) adopted by FinMath (C. Li et al., 2022)

FinMath (C. Li et al., 2022) further improved TagOp (Zhu et al., 2021) by replacing its operator classification layer with autoregressive sequence-to-tree model (Xie & Sun, 2019) to introduce multi step numerical reasoning. Taking flattened question and table as input, FinMath consists of RoBERTa-based (Y. Liu et al., 2019) sequence tagging model and tree-structured neural model that outputs expression tree. The flow of tree-structured neural model is illustrated in **Figure 10**.

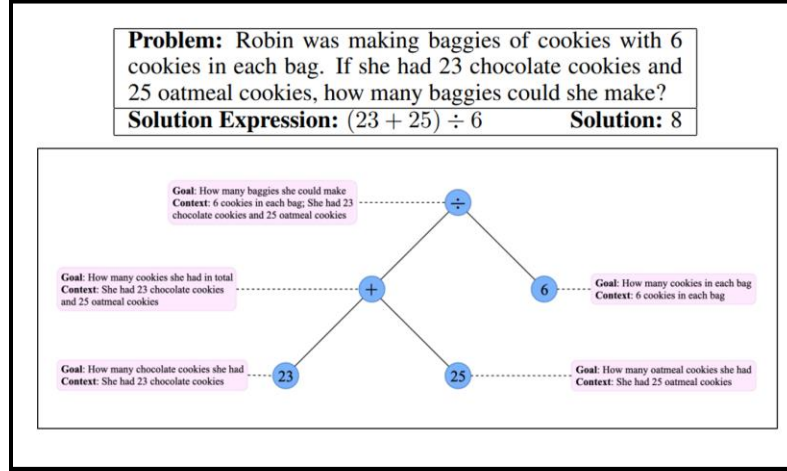


Figure 11 Goal decomposition approach in tree-structured neural model (Xie & Sun, 2019)

As the name implied, the tree-structured neural model works by modelling tree-structure relationship between numbers (Xie & Sun, 2019), in this context the retrieved numbers from input table relevant to answer the given question. To do this, the model first initializes a goal vector representing the problem or calculation based on the question asked, and subsequently obtaining relevant information to solve the question as context vector (Xie & Sun, 2019). Based on the goal vector and context vector, the next token is predicted, which can be number or operator. If the next token predicted is number, then no further decomposition is required because the calculation can be performed. However, if the next token predicted is an operator, this means that new sub-goals (stands for left number operand and right number operand) need to be generated. It will recursively continue until no further decomposition is required. **Figure 11** illustrates this goal decomposition approach in tree-structured neural model by (Xie & Sun, 2019).

2.1.4. Graph Representation of Table Structure

Several works considered table linearization as oversimplification of data structure. Therefore, several works proposed that the tabular structure can be retained as a graph format (S. Li et al., 2024; Nararatwong et al., 2022a; Wang et al., 2022).

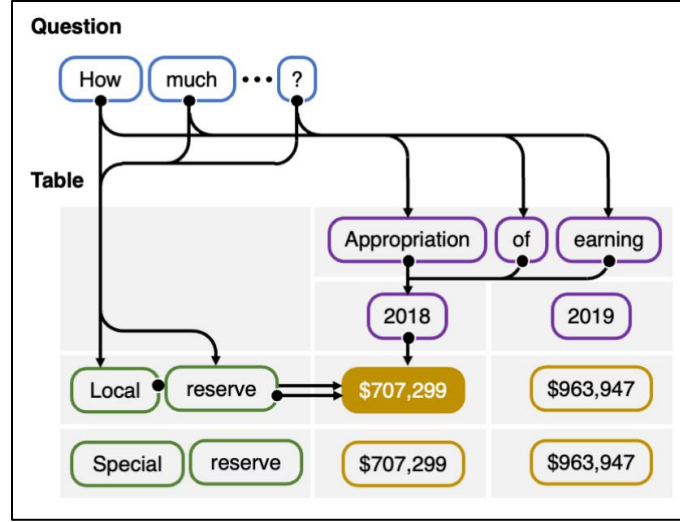


Figure 12 Tabular graph based on GANO (Nararatwong et al 2022)

GANO (Nararatwong et al., 2022a) implemented Graph Neural Network (GNN) to model each cell to their corresponding headers, therefore produces tabular graphs. The tabular graphs produced provide links between questions, table headers, and each cell. Tabular graph based on GANO (Nararatwong et al., 2022a) is illustrated in **Figure 12**.

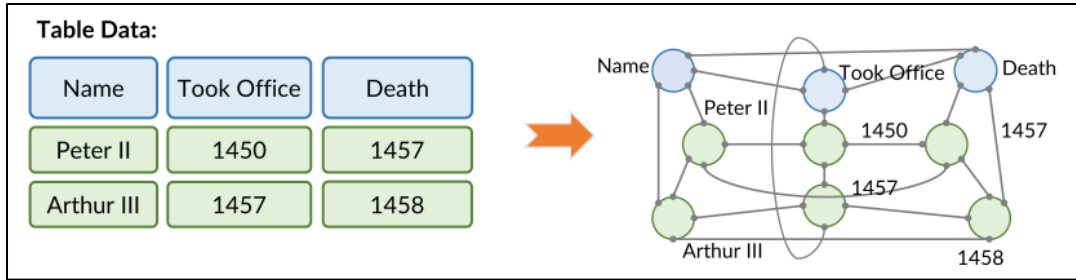


Figure 13 Tabular graph based on Lattice (Wang et al., 2022) and UniD2T (S. Li et al., 2024), linking each cell to the respective header(s), also linking cells in the same row and column

Meanwhile, Lattice (Wang et al., 2022) and UniD2T (S. Li et al., 2024) takes different approach, particularly in terms of the graph produced. Not only linking each cell to the respective header(s), cells that belong to the same column or row are also linked (illustrated in **Figure 13**). Both use T5 (Raffel et al., 2020) as base models, which inherently is text-to-text models. To make the model graph-structure aware, Lattice (Wang et al., 2022) and UniD2T (S. Li et al., 2024) then replaces the original self-attention mask in Transformer block with two new matrices i.e. position matrix and attention-matrix to encode graph-structure. These changes allow for the emergence of structure-aware pre-trained models, with UniD2T (S. Li et al., 2024) taking a more general stance by unifying a more general type of data-to-text task including graph and key value pair as graph-to-text generation tasks.

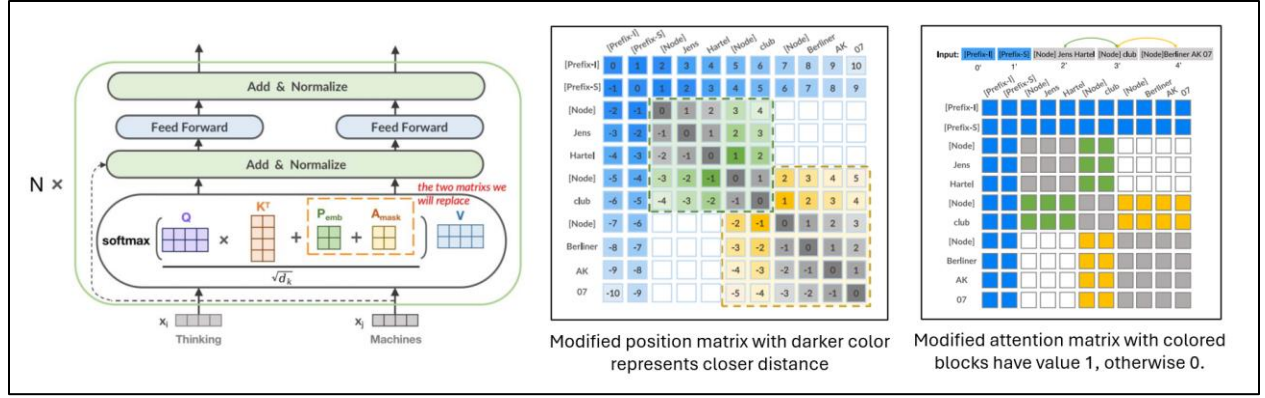


Figure 14 Modified Transformer block in position and attention matrices based on UniD2T (S. Li et al., 2024) and Lattice (Wang et al., 2022)

The change in self-attention matrix is important because self-attention mechanism in the original Transformer architecture cannot effectively capture tabular structure, particularly semantic relationship among cells reflected by non-linear layout structure of a table (Wang et al., 2022). Otherwise, using original Transformer self-attention mechanism, the input graph will be perceived as fully interconnected graph (S. Li et al., 2024). In the other hand, the change to Transformer's positional encoding is to avoid positional bias among distinct cells, by assigning to cells' tokens that share the same row or column, the same relative position (S. Li et al., 2024; Wang et al., 2022). This modification to attention matrix and position matrix is illustrated in **Figure 14**.

The main limitation for UniD2T approach is related to the numerical reasoning (S. Li et al., 2024; Wang et al., 2022), particularly when compared to its alternative in this approach i.e. multi-task learning that inherently designed to handle numerical reasoning (Nararatwong et al., 2022a; Zhu et al., 2021). Multi-task learning is therefore favorable for this approach. However, since it is basically classification task, adjustment might be required in the output. Another notable limitation, as disclosed in Uni2DT paper (S. Li et al., 2024), is the output produced has lower diversity score when compared to the zero-shot output using ChatGPT.

2.1.5. Visual Recognition

Several attempts have tried a different perspective by offering a unique approach that frames table-to-text generation as visual recognition. As also explored by Deng et al (2024) in the context of multimodal Large Language Model, representing table as images or text has comparable performance.

State of the art model in this approach, PixT3 (Alonso et al., 2024), offered pixel-based model for table-to-text generation based on Vision Transformer architecture (Dosovitskiy et al., 2021), showing excellency both in open-ended and controlled scenario using ToTTo (Parikh et al., 2020) and Logic2Text (Chen et al., 2020) dataset. According to the paper, this model's performance surpasses multimodal LLM, i.e. LLaVa (H. Liu et al., 2023) in the original version not fine-tuned for table-to-text generation.

However, visual recognition-based model still struggles with generating texts beyond superficial facts that requires a more in-depth analysis and reasoning process (Alonso et al., 2024), in which the authors proposed a combination of visual recognition and content planning approach.

Another limitation to this approach is related to the input dimension that affects model performance. PixT3 (Alonso et al., 2024) authors disclosed how 16% of tables from ToTTo dataset is too big to be represented by the model effectively.

2.1.6. Language Model Fine-Tuning

With the emergence of Large Language Model, recently, several papers utilized pretrained large language model for table-to-text generation. In this approach, tables are represented as text (linearized).

For example, Yanagimoto et al (2024) fine-tuned small-parameter LLM, OPT-350m, using ToTTo dataset converted into JSON table imbued in alpaca format. Another work i.e. TableLlama (Zhang et al., 2024) is produced using similar approach of fine-tuning Llama2 using annotated alpaca-formatted instruction tuning dataset, but with larger size aiming to serve a more diverse table-related downstream tasks i.e. table interpretation, table augmentation, question answering, fact verification, dialogue generation, and data to text.

This approach may still struggle with hallucination problems, which indicates that the model still uses knowledge acquired during pre-training phase, and also classic problem with inaccurate number displayed in generated content (Yanagimoto et al., 2024).

2.1.7. SQL Query Generation

Other research approached natural language generation task from structured data by aiming to generate SQL query to answer given questions. This generated SQL query is then executed separately to generate final answer.

Works on this approach include Seq2SQL (Zhong et al., 2017), the work which also contributes the dataset WikiSQL for natural language to SQL task, implementing seq2seq architecture to generate SQL query based on user question then execute it to produce an answer. Subsequent work, i.e. SQLova (Hwang et al., 2019) improves the previous work by using two-layers architecture, i.e. BERT-based (Devlin et al., 2019) table-aware encoding layer and LSTM-based NL2SQL layer. A recent work, RAVEN (Theuma & Shareghi, 2024), uses WikiSQL dataset to finetune pretrained language model to generate SQL query for answering a question.

2.2. Dataset Construction for Table-to-text Generation

To implement table-to-text generation for notes to government financial statements, construction of domain-specific dataset with contextual information is required. To introduce numerical reasoning ability to language model, the dataset created also needs to contain the steps of numerical operations required. There are many popular datasets from various domains available for table-to-text generation tasks. **Table 1** summarizes the components of those datasets.

For table-to-text generation, table description and gold facts as target need to be provided in the dataset. Some datasets which focused on table summarization contain highlighted cells to guide

text generation i.e. ToTTo (Parikh et al., 2020) and CTRLSciTab (Guo et al., 2024). This approach is followed by (Chapman et al., 2022), using content-highlighted private dataset to generate financial statements.

Table 1 Popular Datasets for Table-to-text Generation

Dataset	Dataset Component				Dataset Domain
	Table Content	Table Description	Target (Gold Facts)	Reasoning Steps	
WikiSQL (Zhong et al., 2017)	Not Highlighted	✓	✓		General
CTRLSciTab (Guo et al., 2024)	Highlighted	✓	✓		Scientific
NumericNLG (Suadaa et al., 2021)	Not Highlighted	✓	✓		Scientific
ToTTo (Parikh et al., 2020)	Highlighted	✓	✓		General
TableInstruct (Zhang et al., 2024)	Not highlighted	✓	✓		General
Logic2Text (Chen et al., 2020)	Not Highlighted	✓	✓	✓	General
TAT-QA (Zhu et al., 2021)	Not Highlighted	✓	✓	✓	Finance
FinQA (Chen et al., 2021)	Not Highlighted	✓	✓	✓	Finance

Among those datasets, there are three datasets that contain reasoning steps:

- Logic2Text (Chen et al., 2020) dataset that contains table and gold fact as target, which in order to output the golden text complex reasoning over the table is required.
- FinQA (Chen et al., 2021) – domain specific in Finance. Step by step reasoning is provided as a sequence of mathematical and table aggregation operators i.e. addition, multiplication, division, comparison, exponential, maximum, minimum, summation and average. For each table, the questions and reasoning process required to answer the questions are obtained from paid annotators.
- TAT-QA (Zhu et al., 2021) – domain specific in finance. Numerical reasoning is provided as mathematical expression, in which several papers have tried to generate the expressions using various approaches (C. Li et al., 2022; Nararatwong et al., 2022a; Zhu et al., 2021)

It is also worth noting that among the available datasets for table-to-text generation, there is no dataset specific for the domain of public finance. This domain-specific dataset is needed to introduce contextual information for table-to-text generation for notes to government financial statements. This includes e.g. numerous macroeconomic indicators, expenditure budget allocation, and debt financing. Therefore, datasets constructed for this research, particularly covering the domain of public finance, can serve as a contribution to the publicly available dataset for table-to-text generation.

3. Summary of State-of-the-art

Table 2 summarizes the review of previous works related to various approaches in table-to-text generation. Particularly highlighting different ways of representing table in language model and how each work handles numerical reasoning in table-to-text generation.

Table 2 Summary of Previous Works

Previous Work	Table Representation				Numeric Reasoning		
	Linearized	Graph	Image	Table	Retrieval	Single Operation	Multi-step Operation
Highlighted table approach (Chapman et al., 2022; Guo et al., 2024; Parikh et al., 2020)	✓				✓		
Content Planning Approach (Murakami et al., 2017; Puduppully et al., 2022; Puduppully & Lapata, 2021)	✓				✓		
Precomputed Table: (Suadaa et al., 2021)	✓					✓	
Program Generator: TagOp (Zhu et al., 2021), KIQA (Nararatwong et al., 2022b)	✓					✓	
Multi-step operation using tree-structure: FinMath (C. Li et al., 2022)	✓						✓
Multi-step operation using sequential operation: FinQANet (Chen et al., 2021)	✓						✓
Program Generator and Tabular Graph: GANO (Nararatwong et al., 2022a)		✓				✓	
Tabular Graph: Lattice (Wang et al., 2022) UniD2T (S. Li et al., 2024)		✓			✓		
Image representation: PixT3 (Alonso et al., 2024)			✓		✓		
LLM Fine-tuning (Yanagimoto et al., 2024; Zhang et al., 2024)	✓				✓		
Text-to-SQL (Hwang et al., 2019; Theuma & Shareghi, 2024; Zhong et al., 2017)				✓	✓		

Based on the literature review, previous works have various ways to represent table for the language model to generate text based on it. Table linearization still becomes the mainstream way of representing table to the language model (Chapman et al., 2022; C. Li et al., 2022; Murakami et al., 2017; Parikh et al., 2020; Puduppully et al., 2022; Puduppully & Lapata, 2021; Suadaa et al., 2021; Zhu et al., 2021), even though Li et al (2024) argue that linear representation is an oversimplification of data structure. A unique approach is PixT3 which represents tables as images (Alonso et al., 2024), even though it is limited by table dimension that can effectively be represented. Several approaches then consider tabular graph as a way to retain structural representation of table, while it can be effectively encoded using language model (S. Li et al., 2024; Nararatwong et al., 2022a; Wang et al., 2022).

Based on the numerical reasoning capability, the literature review shows that several table-to-text generation works are focused on table summarization (Alonso et al., 2024; Chapman et al., 2022; Parikh et al., 2020; Puduppully et al., 2022; Puduppully & Lapata, 2021; Yanagimoto et al., 2024) or retrieve relevant facts based on given query (S. Li et al., 2024; Wang et al., 2022) therefore having limited reasoning capability, particularly numerical reasoning. However, there are also works that attempted to generate numerical reasoning. Several works use precomputation to handle reasoning (Murakami et al., 2017; Suadaa et al., 2021). Other works use multi head classification to generate numerical reasoning required based on the query (C. Li et al., 2022; Nararatwong et al., 2022a; Zhu et al., 2021) which appears to be a promising approach. Particularly FinMath (C. Li et al., 2022) which implements tree-structured numerical reasoning that enables multi-step numerical reasoning, and GANO (Nararatwong et al., 2022a) that utilize GNN and tabular graph for evidence extraction.

From the previous works, it appears that combining the implementation of tabular graph representation (S. Li et al., 2024; Nararatwong et al., 2022a; Wang et al., 2022) and tree-structured numerical reasoning that enables multi-step numerical reasoning (C. Li et al., 2022) in table-to-text generation has not been researched. Particularly whether this combined approach can improve model performance. Therefore, this research aims to improve the model performance from previous works by representing the table as graph based on Lattice (Wang et al., 2022) and UniD2T (S. Li et al., 2024). Also, improving numerical reasoning by combining GNN-based evidence extractor (Nararatwong et al., 2022a) and tree-structured multi-step numerical reasoning approach (C. Li et al., 2022; Xie & Sun, 2019). The research question proposed is as follows:

- Can graph-based tabular representation and multi-step numerical reasoning improve the accuracy of numerical reasoning to tabular financial data for notes to financial statement generation?

Aside from methodology enhancement proposed, this research also aims to contribute table-to-text generation dataset, specifically for public finance domain. This dataset also contains numerical reasoning which follows the format of previous datasets i.e. FinQA (Chen et al., 2021) and TAT-QA (Zhu et al., 2021).

Activity	Schedule		March				April				May			
	Start	End	1	2	3	4	1	2	3	4	1	2	3	4
1. Dataset Construction	3-Mar-25	11-Apr-25												
1.1. Data Source Acquisition	3-Mar-25	4-Mar-25												
1.2. Table and Golden Text Extraction	5-Mar-25	14-Mar-25												
1.3. Task and Reasoning Annotation	17-Mar-25	11-Apr-25												
2. Model Development	10-Mar-25	9-May-25												
2.1. Architecture Coding	10-Mar-25	28-Mar-25												
2.2. Model Training	14-Apr-25	25-Apr-25												
2.3. Model Evaluation	28-Apr-25	9-May-25												
3. Solution Prototype Development	11-May-25	30-May-25												
3.1. Solution Prototype Coding	11-May-25	23-May-25												
3.2. Solution Prototype Hosting	25-May-25	30-May-25												
4. Reporting, Publication and Presentation	14-Apr-25	30-May-25												
4.1. Report Writing	14-Apr-25	16-May-25												
4.2. Paper Publication	19-May-25	30-May-25												

Figure 15 Research project timeline

4. Research Project Plan

In order to implement a model that is able to generate text with numerical reasoning capability, this research plans to improve TagOp model (Zhu et al., 2021) by combining FinMath’s numerical expression tree (C. Li et al., 2022) and graph representation of tabular data based on GANO (Nararatwong et al., 2022a) and UniD2T (S. Li et al., 2024; Wang et al., 2022) in the context of notes to financial statement generation using Indonesia government financial data and Indonesia macroeconomic data.

This research project consists of 4 main activities as outlined in **Figure 13**. The project is planned to be executed in 3 months starting from March 3, 2025, until May 30, 2025.

4.1. Dataset Construction

To generate notes to government financial statements using table-to-text generation, two types of data are needed. Tabular data as the source of notes to financial statements and textual data from the report as golden or target text. The required tabular data and corresponding source are listed in **Appendix B**.

For the golden text, original sentences from central government financial statements (LKPP) for 2 fiscal years (2022 and 2023) are used. To provide additional golden texts, particularly related to macroeconomic analysis, monthly report of macroeconomic situation released by Ministry of Finance Fiscal Policy Agency for the fiscal year 2024 and 2023 are used. It is expected to obtain 1000 observations (golden texts) from these reports. For numerical reasoning, manual annotation will be conducted based on the golden text and the corresponding input table. The annotation format follows TAT-QA dataset (Zhu et al., 2021).

The dataset constructed in this step becomes a contribution for two reasons. First, it provides table-to-text dataset specialized in public finance domain which includes macroeconomic analysis for generating notes to government financial statements. Second, the dataset contains numerical reasoning over table, also complemented with contextual information underlying the reasoning steps.

4.2. Model Development

The proposed model is based on TagOp (Zhu et al., 2021) which consists of multi classifier i.e. evidence extractor, operator classifier, and scale classifier. Evidence extractor obtains supporting evidence from the provided table input based on sequence tagging. Sequence tagging assigns to each token “I” or “O” label, with tokens labeled with “I” are the supporting evidence to be used in the next step to produce the answer. As explained by Zhu et al (2021), the probability of tagging each token in the evidence extractor is calculated using softmax on 2-layered feed-forward neural network using GELU as activation function on sub token h_t as:

$$P_t^{tag} = \text{softmax}(FFN(h_t))$$

The operator classifier applies multi-class classification to predict the correct operator from the following options: spans, sum, count, average, multiplication, division, difference, and change ratio

(Zhu et al., 2021). Similar to sequence tagging, the probability of operator is calculated using softmax on 2-layered feed-forward neural network using GELU as activation function. However, [CLS] token, which represents sentence-level BERT classification, is used in this probability calculation.

$$P^{op} = \text{softmax}(FFN([CLS]))$$

The scale classifier handles the prediction of table value scaling. It takes [CLS] token and table vector representation and calculates probability using softmax on 2-layered feed-forward neural network using GELU as activation function.

$$P^{scale} = \text{softmax}(FFN([CLS]; h_{tab}))$$

Subsequent research has modified this basic architecture. FinMath (C. Li et al., 2022) instead of using operator classifier, enhance it by replacing the operator classifier with tree-structured neural model that outputs expression tree. This modification allows for multi-step numerical operation compared to the basic architecture that only outputs a single numerical operation.

Another modification is also conducted by (Nararatwong et al., 2022a) which added tabular graph and GNN layer in the evidence extractor. Tabular graph allows for better representation of table structure compared to the basic architecture that represents the input table as sequence of flattened table.

This research combines these two enhancements on the original TagOp model, i.e. tabular graph representation of input table and tree-structured neural model for numerical reasoning, as visualized in **Figure 16**.

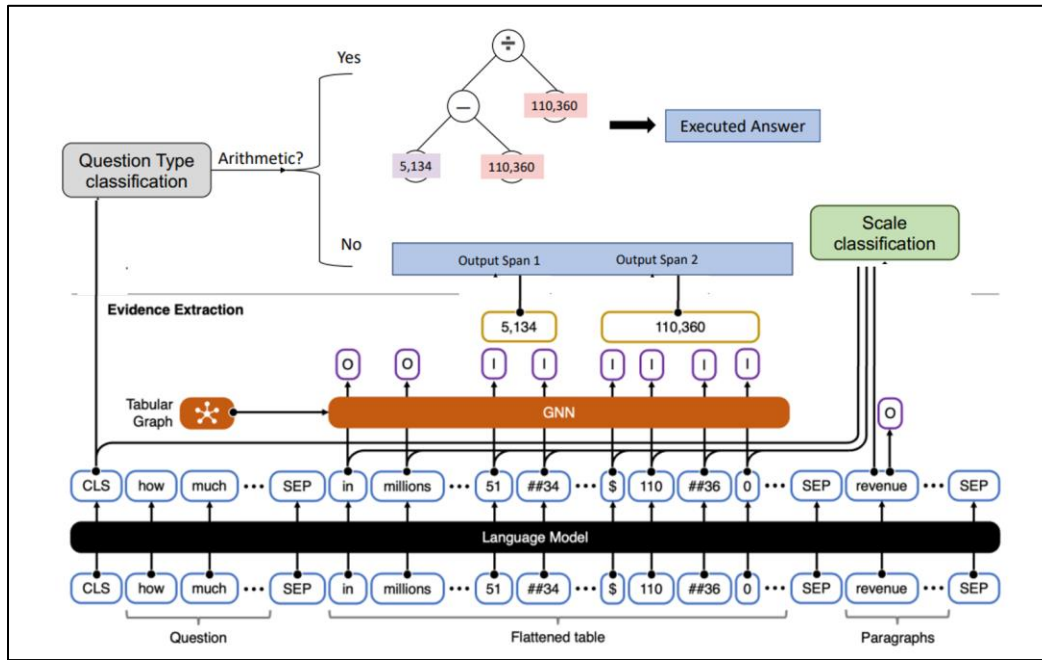


Figure 16 The proposed architecture for table-to-text based on TagOp model

This research proposes additional modification with regards to the tabular graph design. The original tabular graph based on GANO (Nararatwong et al., 2022a) linking each tokens in the question through the table row and column header to each particular cell in the table, as previously illustrated in **Figure 10**.

Graph representation of input table is implemented based on Lattice and UniD2T (S. Li et al., 2024; Wang et al., 2022). In this approach, tabular data is transformed into levi graph i.e. unlabeled bipartite graph. In this graph, each cell that belongs in the same row is given forward edge and corresponding reverse edge. The same thing applies to cells that belong in the same column, forward and reverse edge are introduced. **Figure 15** illustrates this proposed modified tabular graph structure.

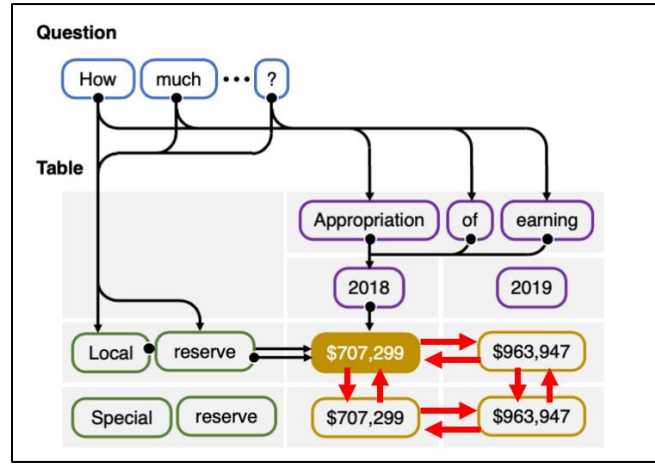


Figure 17 Modified tabular graph structure proposed

For optimization, based on TagOp (Zhu et al., 2021), the overall loss to minimize during training is a combination of losses (negative log likelihood) from the classification tasks in the model as follows:

$$\mathcal{L} = \text{NLL}(\log(P^{tag}), G^{tag}) + \text{NLL}(\log(P^{op}), G^{op}) + \text{NLL}(\log(P^{scale}), G^{scale})$$

Which G represents the ground truth based on the training dataset prepared. However, number order classifier from the original implementation is not used, and is replaced by expression tree based on FinMath (C. Li et al., 2022).

Then, to answer the research question whether this approach can improve model performance, this research will compare the performance of the proposed model with TagOp (Zhu et al., 2021), FinMath (C. Li et al., 2022) and GANO (Nararatwong et al., 2022a). For this performance evaluation between the models, I followed FinMath (C. Li et al., 2022) that used exact match (EM) and numeric-focused F1 score.

4.3. Prototype Development of Solution

Since the goal of the research is to produce notes to financial statements, a prototype for the financial statement generator will be provided. This solution arranges the parts of financial statements based on the order, and for each part, the model will output texts based on the arranged

relevant questions and input table corresponding to each part of financial statement. Python-based web application framework will be used to develop the solution prototype.

4.4. Reporting, Publication and Presentation

The reporting of this research is planned to be conducted starting from the model training phase. Particularly to elaborate the experimental setup e.g. computational budget (including GPU hours), computation infrastructure details, and hyperparameter settings.

The research report is planned to be completed on May 16, to meet the earliest deadline for the target conference venue. There are two conference venues that can be considered suitable for this research plan:

- **ACM International Conference on AI in Finance 2025.** This conference is scheduled to be held around November 2025. Currently there is no further information related to the deadline for submission in this venue.
- **Empirical Methods in Natural Language Processing (EMNLP) 2025.** Especially because the main theme for this year is interdisciplinary recontextualization of NLP. The conference will be held in November 2025, with the deadline for submission on 19 May 2025.

Finally, a poster presentation for this research is scheduled to be held around August or September 2025.

5. Conclusion

The current development of Language Model that allows for various text-generation tasks brings the potential of automating the construction of government notes to financial statement. Many of the data sources are in tabular format, therefore table-to-text generation downstream task may be suitable. However, the complex task in writing notes to financial statements, beyond accurate summarization and explanation of certain facts from a table, is when the explanation requires numerical and comparative reasoning. Several works have attempted to introduce numerical reasoning in table-to-text generation tasks by language models, particularly through structural representation of table and prediction of correct numerical expression. This research proposes an improvement to TagOp model (Zhu et al., 2021) by combining FinMath’s numerical expression tree (C. Li et al., 2022; Xie & Sun, 2019) and graph representation of tabular data based on GANO (Nararatwong et al., 2022a) and UniD2T (S. Li et al., 2024; Wang et al., 2022) in the context of notes to financial statement generation using Indonesia government financial data and Indonesia macroeconomic data.

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Appendix A – Dataset Sources

Data	Source
1. Indonesia GDP based on business sector (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/NjUjMg==/-seri-2010--pdb-menurut-lapangan-usaha-seri-2010--milyar-rupiah-.html
2. Indonesia GDP based on Expenditure (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MTk1NiMy/-seri-2010--2--pdb-triwulanan-atas-dasar-harga-konstan-menurut-pengeluaran--milyar-rupiah-.html
3. Global Commodity Prices (World Bank)	https://www.worldbank.org/en/research/commodity-markets
4. Global Economic Prospects (World Bank)	https://www.worldbank.org/en/publication/global-economic-prospects#downloads
5. World Economic Outlook (IMF)	https://www.imf.org/en/Publications/WEO
6. OECD Economic Outlook	https://www.oecd.org/en/publications/serials/oecd-economic-outlook_g1ghgh13.html
7. Inflation Data	https://www.bps.go.id/id/statistics-table/1/OTA4IzE=/inflasi-umum--inti--harga-yang-ditatur-pemerintah--dan-barang-bergejolak-inflasi-indonesia--2009-2024.html
8. 10Y Bond Yield	https://id.investing.com/rates-bonds/indonesia-10-year-bond-yield-historical-data
9. Currency Exchange Rate	https://id.investing.com/currencies/usd-idr-historical-data
10. Export (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MTc1MyMy/nilai-ekspor-migas-nonmigas--juta-us--.html
11. Import (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MTc1NCMy/nilai-impor-migas-nonmigas--juta-us--.html
12. Trade Balance (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/NDk4IzI=/nilai-neraca-perdagangan.html
13. Balance of Payments (BI)	https://www.bi.go.id/id/statistik/ekonomi-keuangan/seki/default.aspx#headingFour
14. Foreign Exchange Reserves (BI)	https://www.bi.go.id/id/statistik/ekonomi-keuangan/seki/default.aspx#headingFour
15. BI 7 Days repo rate (Statistics Indonesia)	https://www.bps.go.id/en/statistics-table/2/Mzc5IzI=/bi-rate.html
16. Monetary Operation (BI)	https://www.bi.go.id/id/statistik/ekonomi-keuangan/seki/default.aspx#headingOne
17. Active Circulation of Money (BI)	https://www.bi.go.id/id/statistik/ekonomi-keuangan/seki/default.aspx#headingOne
18. IDX Composite (YFinance API)	https://finance.yahoo.com/quote/%5EJKSE/history/
19. Gini Ratio (Statistics Indonesia)	https://www.bps.go.id/en/statistics-table/2/OTgjMg==/gini-ratio-menurut-provinsi-dan-daerah.html

20. Distribution of Per Capita Expenditure Allocation (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/1/OTQ2IzE=/distribusi-pembagian-pengeluaran-per-kapita-dan-indeks-gini--2010-2017.ht
21. Farmer exchange rate (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MTcxNyMy/ntp--nilai-tukar-petani--menurut-sektor--2018-100-.html
22. Fisherman exchange rate (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MTcxMyMy/ntn--nilai-tukar-nelayan--menurut-subsektor--2018-100-.html
23. Human Development Index (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/NDk0IzI=/-metode-baru-indeks-pembangunan-manusia-menurut-provinsi.html
24. Poverty level (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MTkylzI=/persentase-penduduk-miskin--p0--menurut-provinsi-dan-daerah.html
25. Unemployment Rate (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/NTQzIzI=/tingkat-pengangguran-terbuka-menurut-provinsi.html
26. Workforce Participation Level (Statistics Indonesia)	https://www.bps.go.id/id/statistics-table/2/MjIwMCMY/tingkat-partisipasi-angkatan-kerja-menurut-jenis-kelamin.html
27. Purchasing Manager Index (Trading Economics)	https://tradingeconomics.com/indonesia/manufacturing-pmi
28. Indonesia Crude Price (ICP)	https://migas.esdm.go.id/post/harga-minyak-mentah
29. Central Government Financial Statement	https://djpb.kemenkeu.go.id/portal/id/laporan-keuangan-pemerintah-pusat-lkpp.html
30. Monthly Economic and Finance Report Fiscal Policy Agency	https://fiskal.kemenkeu.go.id/analisis/laporan-ekonomi-dan-keuangan-bulanan

Part 2

Research Paper

IndoMPF: Dataset and Tabular-Graph Numerical Reasoning Baseline Model for Generating Notes to Government Financial Statements

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Abstract

Manually drafting Notes to Government Financial Statements is a time-intensive process prone to errors, requiring complex numerical and contextual reasoning that challenges existing language models. This challenge is compounded by the lack of specialized datasets, particularly for the domain of Indonesian public finance. To address this gap, we introduce two key contributions. First, we present the IndoMPF Dataset, a novel table question-answering dataset focused on Indonesian macroeconomic and public finance, specifically designed with questions requiring multi-step numerical reasoning. Second, we introduce a general pipeline design for generating notes to government financial statement from tabular data. This pipeline include table retrieval model, baseline table question-answering with multi-step numerical reasoning model, and text generation model. Our experiments show that fine-tuning an embedding model with Multiple Negative Ranking loss achieves 97% accuracy in the table retrieval sub-task. While our graph-aware baseline table question-answering model (IndoMPF-GATe) demonstrated the task’s architectural challenges, our extensive benchmarks of open-source LLMs reveal that Qwen3-8B is a top performer, achieving 54.86% correct answer in a zero-shot setting. Crucially, our analysis shows all models struggle with increasing table size and reasoning complexity, highlighting key areas for future research. Finally, we demonstrate that instruction-tuned models can successfully convert these question-answer pairs into coherent financial notes with high semantic fidelity. This work establishes a foundational dataset and a comprehensive set of benchmarks to spur further research in automating complex government financial reporting. The dataset and code are publicly available ¹.

¹Project GitHub page: <https://github.com/mihoku/IndoMPF>. Explore IndoMPF dataset content in the webpage <https://rezadr.pythonanywhere.com/>

1 Introduction

Notes to financial statements accompany a financial statement to enhance its understandability and to avoid misunderstanding. Notes to financial statements from private entities commonly consist of general information about reporting entity, accounting policy, and explanation of each financial statement post. Government financial statements, on the other hand, embody government responsibility to public financial management. Therefore, particularly in Indonesia, as required in the Government Accounting Standard issued by Committee of Government Accounting Standard (KSAP), there are two extra components to the notes, i.e. budget information, summary of fiscal/monetary policy and macroeconomic situation.

The process of manually drafting notes to financial statements is time-intensive. It is also prone to content inaccuracy and incompleteness of information disclosure, which requires additional control process e.g. financial statement review by government internal auditor unit. Meanwhile, the current development of Language Model that allows for various text-generation tasks brings the potential of automating this business process. Numerous data sources in tabular format are required to construct notes to government financial statement. Therefore, table-to-text generation downstream task has potential to solve this problem.

However, automating the generation of government Notes to Government Financial Statement is not a simple summarization task. A significant challenge lies in performing the complex numerical and comparative reasoning required to produce meaningful explanations. Existing language models often struggle with this, facing issues with hallucination and a limited ability to move beyond retrieving superficial facts from a table. For instance, as Example A in **Figure 1** illustrates, explaining a decrease in revenue from natural resources re-

quires a multi-step analysis: first, identifying the constituent revenue subtypes (oil, gas, mining), and second, determining which subtype had the largest year-on-year decrease to frame it as the main contributor.

Example A

B.2.1.2.1. Natural Resources Revenue

Realization of Natural Resources Revenue in FY2023 amounted to Rp254,930,031,073,614 or reached 113.83 percent of the amount set in the State Budget of Rp223,812,141,813,000. Natural Resources Revenue in FY2023 was Rp14,567,965,197,170 smaller or 1.42 percent lower compared to the realization in FY2022 which amounted to Rp269,370,896,270,384 which was mainly due to 2 decreases in oil and mineral gas revenue.

A

Account Code	Account Description	Budget	Budget Realization 2023	% of Budget	Budget Realization 2022	Increase/Decrease	% Increase/Decrease
411	REVENUE FROM NATURAL RESOURCES	253,313,141,813.000	239,369,256,794.94	94.53%	14,567,965,197.17	-14,567,965,197.17	-14.24%
4111	REVENUE FROM MINERAL GAS	76,761,123,000.000	67,403,000.000	0.88%	11,191,000,000.00	-11,191,000,000.00	-100.00%
4112	REVENUE FROM PETROLEUM GAS	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4113	REVENUE FROM OIL AND COAL REVENUE	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4114	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4115	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4116	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4117	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4118	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4119	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4120	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4121	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4122	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4123	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4124	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4125	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4126	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4127	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4128	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4129	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4130	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4131	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4132	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4133	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4134	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4135	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4136	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4137	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4138	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4139	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%
4140	REVENUE FROM OTHER MINERAL RESOURCES	176,552,000.000	176,552,000.000	100.00%	176,552,000.000	0.000	0.00%

Task A: Summarizing a row in the table

Task B: Explanation based on the value from other rows

Example B

Central Government Spending by Organization/Budget Division

The realization of Central Government Expenditure in FY2023 according to the Budget Section was the largest in 999 (State Treasurer) amounting to Rp1,087,557,384,455,403 or 48.56 percent of the total Central Government Expenditure. Meanwhile, the total realization of Central Government Expenditure in FY2023 was Rp2,230,456,800,897 (other than State Treasurer) was Rp1,152,229,317,043,761. Of the total realization of expenditure, the largest realization of expenditure in Ministries/Institutions was in the Ministry of PUPRI amounting to Rp174,324,456,800,897 or 15.13 percent of the total realization of Central Government Expenditure for Ministries/Institutions.

To calculate the 15.13 percentage, the expenditure from state treasurer must be excluded from total expenditure

Unit Code	Description	Expenditure Budget	Budget Realization	Budget Deviation	% of Budget
033	Ministry of Public Works and Public Housing	125,211,457,417,000	174,324,456,800,897	49,096,999,329,897	139.22%
999	State Treasurer	1,301,613,134,325,000	1,087,557,384,455,403	-214,055,748,879,597	83.53%
	Total Expenditure	2,302,457,800,414,000	2,230,456,800,897	-72,001,000,503,103	97.28%

Figure 1: Example of numerical and contextual reasoning complexity required in Notes to Financial Statement

Another example is the contextual understanding requirement involved in calculating expenditure percentages as illustrated in Example B of **Figure 1**. This calculation requires exclusion of certain items, in this context the expenditures from the State Treasurer, before performing the calculation. Retrieving the correct cells and identifying the correct sequence of operations is a profound challenge for standard text-generation models.

While various table-to-text methods exist, they often face limitations. First regarding the representation of data structure. Many models rely on table linearization, which can oversimplify complex table structures (Chapman et al., 2022; Li et al., 2022; Murakami et al., 2017; Parikh et al., 2020; Pudupully et al., 2022; Pudupully and Lapata, 2021; Suadaa et al., 2021; Zhu et al., 2021). Regarding table linearization, Li et al. (2024) argue that linear representation is an oversimplification of data structure, and therefore several works propose alternative representation of table structure e.g. graph (Li et al., 2024; Nararatwong et al., 2022; Wang et al., 2022) and image (Alonso et al., 2024).

Numerical reasoning remains a challenge in table-to-text generation task, in which the model proposed struggle with generating texts beyond simple retrieval of superficial facts from the table

(Alonso et al., 2024; Chapman et al., 2022; Guo et al., 2024; Parikh et al., 2020). Furthermore, while some models have incorporated numerical reasoning, they are often limited to single-step operations (Nararatwong et al., 2022; Suadaa et al., 2021; Zhu et al., 2021). Other approaches have endeavored to propose multi-step reasoning (Chen et al., 2021a; Li et al., 2022), although the domain for the task is general finance, which may not be generalizable for public finance and macroeconomic domain.

To address this specific research gap, this work provides two key contributions:

1. **The IndoMPF Dataset:** A new, specialized table question-answering dataset focused on Indonesian macroeconomic and public finance. It is explicitly designed to include questions that require the kind of multi-step numerical reasoning found in real-world government financial reporting.

2. A set of component to form a pipeline for generating notes to government financial statement from tabular data. This pipeline include table retrieval model, baseline table question-answering with multi-step numerical reasoning model, and text generation model. For baseline table question-answering model, we develop Indonesia Macroeconomic and Public Finance Graph Transformer Model (IndoMPF-GATE) model which represents tabular data as a graph to better capture structural information and uses a transformer-based decoder to generate reasoning programs. This establishes a benchmark for this new, complex task.

2 Pipeline Component for Generating Notes to Financial Statement and Research Objective

The design for automatic generation of notes to government financial statement pipeline consists of the following components:

- Each section in financial statement is represented by a set of questions.
- Embeddings of information regarding all available source tables are stored in vector database.
- Based on given question for each section, retrieve relevant table from vector database using

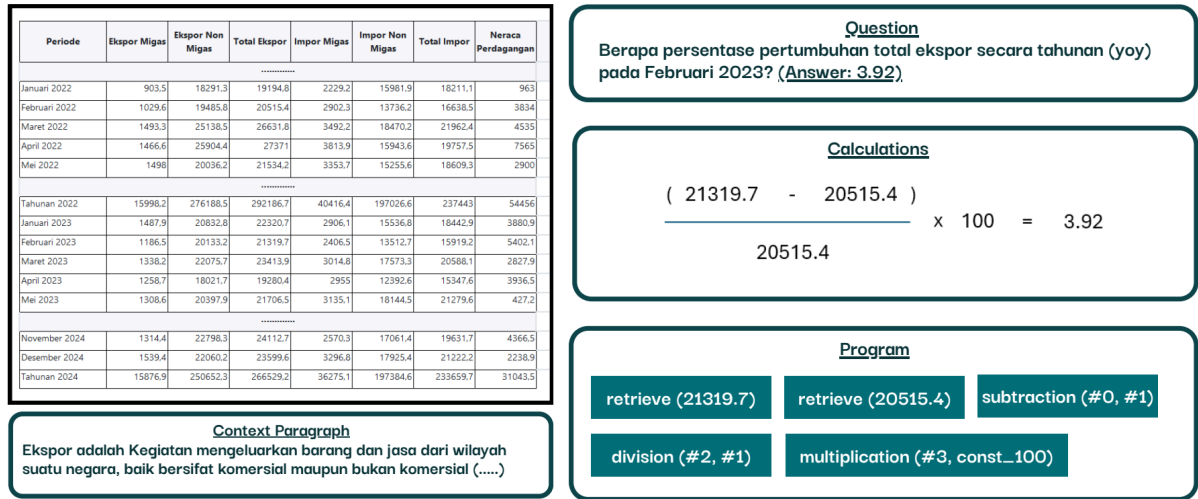


Figure 2: IndoMPF Table Question Answering Dataset Preview

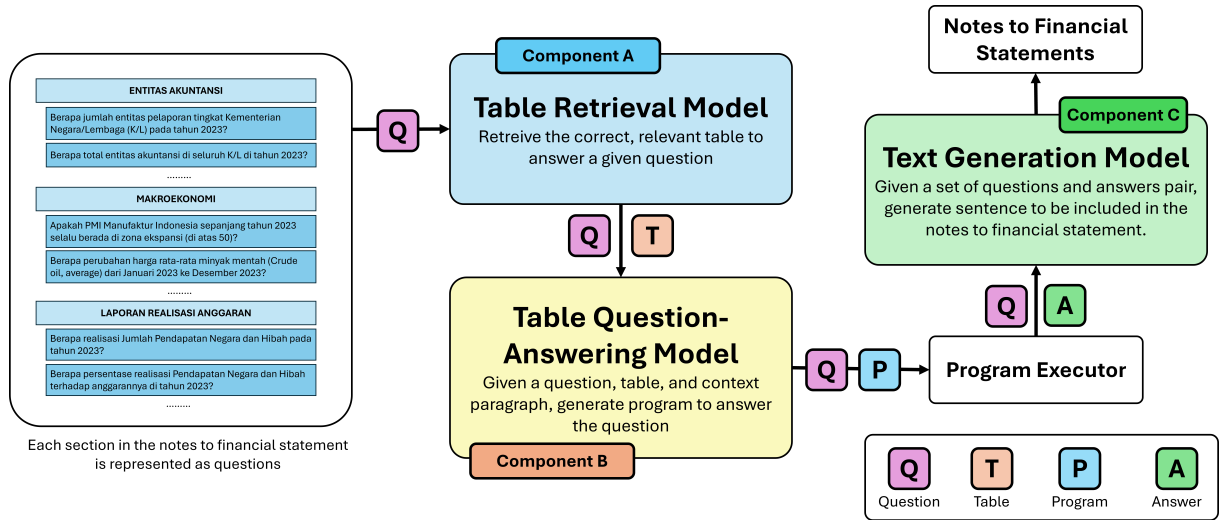


Figure 3: Pipeline Design to Automate the Generation of Notes to Financial Statement

ing retrieval model (Component A of **Figure 3**).

- Each pair of question and table retrieved are passed as input to the table question-answering model to generate a program to answer the question (Component B of **Figure 3**).
- Program executor will use the program to get the answer to the question.
- Answered question are used as prompt context to generate each sentence in government notes to financial statement using text-generation model (Component C of **Figure 3**).

This research aims to prepare the necessary components to build the pipeline for generation of notes

to financial statement:

- Construction of dataset, i.e. reference text dataset and table question answering dataset with numerical reasoning with specialized domain in macroeconomic and public finance.
- Fine tuning of embedding model using constructed domain-specific dataset for table retrieval task.
- Development of baseline model for table question-answering with numerical reasoning.
- LLM instruction-tuning for text-generation task for notes to financial statement generation.

3 Related Works

3.1 Table Question-Answering Dataset

To implement table-to-text generation for notes to government financial statements, construction of domain-specific dataset with contextual information is required. To introduce numerical reasoning ability to language model, the dataset created also needs to contain the steps of numerical operations required. There are many popular datasets from various domains available for table-to-text generation tasks. **Table 1** summarizes the components of those datasets.

For table-to-text generation, table description and gold facts as target need to be provided in the dataset. Some datasets which focused on table summarization contain highlighted cells to guide text generation i.e. ToTTo (Parikh et al., 2020) and CTRLSciTab (Guo et al., 2024). This approach is followed by Chapman et al. (2022), using content-highlighted private dataset to generate financial statements.

Among those datasets, there are three table question answering datasets in finance domain that contain reasoning steps:

- TAT-QA (Zhu et al., 2021) – domain specific in finance. Numerical reasoning is provided as mathematical expression.
- MultiHiertt (Zhao et al., 2022) – domain specific in Finance. Similar to TAT-QA, numerical reasoning is provided as mathematical expression.
- FinQA (Chen et al., 2021b) – domain specific in Finance. Step by step reasoning is provided as a sequence of mathematical and table aggregation operators i.e. addition, multiplication, division, comparison, exponential, maximum, minimum, summation and average. For each table, the questions and reasoning process required to answer the questions are obtained from paid annotators.

It is also worth noting that among the available datasets for table-to-text generation, there is no dataset specific for the domain of public finance. This domain-specific dataset is needed to introduce contextual information for table-to-text generation for notes to government financial statements. This includes e.g. numerous macroeconomic indicators, expenditure budget allocation, and debt financing.

Therefore, IndoMPF dataset constructed for this research, particularly covering the domain of public finance, can serve as a contribution to the publicly available dataset for table-to-text generation.

The IndoMPF dataset also significantly raises the bar for financial reasoning tasks in two dimensions: scale and complexity. In terms of scale, IndoMPF dataset features tables up to 1,460 rows, an order of magnitude larger than the 20-30 row tables in datasets like FinQA and TAT-QA. Another primary limitation of existing financial question-answering datasets is their relatively narrow scope of numerical operations, which often restricts them to simple arithmetic. For instance, TAT-QA and FinQA each contain 10 operator types, while MultiHiertt contains only 4. To address this gap, our IndoMPF dataset introduces 37 distinct operators, encompassing not only standard arithmetic but also more complex table-specific functions, i.e. trace, filter, and unique. This expanded operational vocabulary is crucial for modeling the nuanced, multi-step reasoning required to accurately interpret government financial statements. Therefore, this combination creates a robust and challenging benchmark for developing the next generation of financial analysis models.

3.2 Table Question-Answering Model

Table-to-text generation, the task of creating textual descriptions from structured data, has been explored through various approaches. A foundational method involves using highlighted cells to guide text generation. Pioneered by ToTTo (Parikh et al., 2020), this technique linearizes tables and uses highlighted cells as a signal for Transformer-based models to focus on specific data points. This work established that using relevant table subsets and metadata, such as page titles, improves performance. This approach was later adapted by Chapman et al. (2022) for generating financial statement narratives from tabular data, similarly relying on linearized tables with special tokens to denote structure and highlighted content. Another significant strategy is content planning, where the model is guided by a predefined plan of information to include. This can involve using generalization tags for numerical operations (Murakami et al., 2017), employing template-based inputs to bridge the gap between structured data and natural language for models like GPT-2 and T5 (Suadaa et al., 2021), or a two-stage process of first verbalizing salient

Dataset	Numerical Reasoning	Dataset Domain	Oper. Types	Max. Table Dimension	Tables	QA
TAT-QA (Zhu et al., 2021)	Math Expr.	Finance	10	30 rows	2.757	16.552
FinQA (Chen et al., 2021b)	Math Expr.	Finance	10	20 rows	2.789	8.281
MultiHiertt (Zhao et al., 2022)	Multi-step	Finance	4	11 rows (avg)	2.513	10.440
IndoMPF (ours)	Multi-step	Macro econ. & Public Finance	37	1.460 rows	156	7.007

Table 1: Comparison of table question-answering datasets in finance domain

entities and then generating a paragraph-level plan (Puduppully and Lapata, 2021; Puduppully et al., 2022).

To address the need for numerical reasoning, the program generator approach has been developed. These methods generate executable programs to derive answers from tables. For instance, FinQANet (Chen et al., 2021) uses an LSTM decoder to perform multi-step reasoning over linearized tables. TagOp (Zhu et al., 2021) introduced a multi-task learning framework with distinct classifiers for evidence extraction and numerical operations. This architecture inspired several enhancements, such as FinMath (Li et al., 2022), which replaced the operator classification with a more sophisticated sequence-to-tree model for multi-step calculations. Another related research avenue focuses on SQL query generation. Models like Seq2SQL (Zhong et al., 2017), SQLova (Hwang et al., 2019), and RAVEN (Theuma and Shareghi, 2024) are trained to translate natural language questions into SQL queries, which are then executed on the database to retrieve the final answer.

Recognizing that table linearization can oversimplify complex data structures, several studies have proposed using graph representations to preserve the table’s inherent structure. GANO (Nararatwong et al., 2022) models tables as graphs and uses a Graph Neural Network (GNN) to link cells to their corresponding headers. Other models like Lattice (Wang et al., 2022) and UniD2T (Li et al., 2024) create more detailed graphs where cells in the same row or column are also connected. They adapt the Transformer architecture by modifying the self-attention and position encoding mechanisms to make the model "structure-aware," thereby preventing the model from treating the input as a simple, fully connected graph. Despite their strengths in capturing structure, these models can face limitations in numerical reasoning compared to multi-task learning approaches (Nararatwong et al., 2022;

Zhu et al., 2021). Some other works combined graph representation of table and tree-based decoder to imbue numeric reasoning e.g. RegHNT (Lei et al., 2022) and MVGE (Wei et al., 2023). However this approach limits the reasoning capability to arithmetic-based questions.

4 Methodology

4.1 Dataset Construction

A novel dataset is required to train the model for table question-answering task. **Figure 4** illustrates the preprocessing for the dataset.

4.1.1 Table Collection

Next is to provide description paragraph to each table collected. It aims to provide descriptive information about the table for RAG purposes. It consists of:

- General information about the table to provide contextual information about the table, including the definition of terms in the table. The information is obtained from various sources e.g. regulation and official reports.
- Column description, obtained from table meta-data

4.1.2 Reference Text Collection

Aligning to the objective of generating notes to financial statements, reference text are sentences obtained from relevant government reports. The reference texts serve as a valid interpretation of information from the source table. The government reports used for source of reference text are as follows:

- **Central Government Financial Statement (LKPP)** issued by Ministry of Finance Republic of Indonesia. These documents serves as the main source for reference text that aligns with the objective of this research.

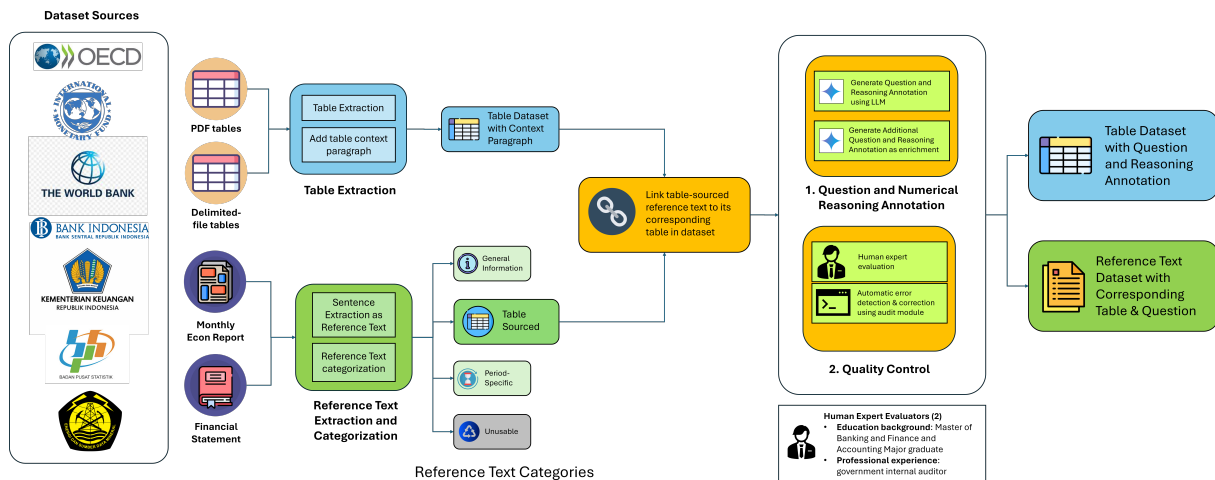


Figure 4: Preprocessing for Dataset

- **Monthly Financial and Economic Report** issued by Ministry of Finance Republic of Indonesia. This is an additional source which inclusion aims to train the model for question answering in macroeconomic analysis.

The reference text (sentences) are then manually annotated for categorization into several groups:

- **Table Answerable** (answerable with table), when the information from the golden text can be obtained from available table. It doesn't necessarily have to contain numbers, but an analysis of increase or decrease, and the affecting factor can also be answered from the table.
- **General Information** (general info to be put in knowledge graph), contains information that is generally relevant for all fiscal year. For instance: general description/definition of certain concept (e.g. "Oil and gas lifting is the volume of oil and gas production from national oil and gas fields that is ready for sale") and general principle regarding the relationship between two concepts (e.g. "Oil and gas lifting is a benchmark in calculating several components in the State Budget, including Non-Tax State Revenue (PNBP) from the oil and gas sector, tax revenue in the oil and gas sector, and regional transfers in the form of Revenue Sharing Transfer Funds (DBH) for oil and gas producing regions").
- **Event-Specific** (event specific information (undecided)), contains information that is only

relevant for a fiscal year, e.g. "In 2023, two new ministries/institutions were consolidated into the LKPP, namely the National Food Agency and the Indonesian Capital Authority."

- **Unusable**, doesn't fall into the above categories.

Manual categorization of fiscal year 2023 reference text is completed using table annotation tool as further described in Appendix A.

For other fiscal years, text-classification is applied to carry out the categorization of reference text, using BERT-based embedding model. Two models are trained i.e. indolem/indobert-base-uncased (Koto et al., 2020), indobenchmark/indobert-large-p1 (Wilie et al., 2020). These models are selected considering the reference text are all in Bahasa Indonesia. The performance of the models after training for 3 epochs can be summarized in Table 2

Due to the inherent class imbalance in the manually annotated dataset, the Macro F1-score was used as the primary metric for a fair comparison of model performance across all categories. Upon evaluation on a held-out test set, both models achieved a nearly identical Macro F1-score of 0.88. Given the comparable performance, the more computationally efficient indobert-base-uncased model was selected for the final task of classifying the remaining reference texts.

Using text classification model selected, the remaining uncategorized reference texts are categorized. The distribution of each reference text with

Metric	Base	Large
<i>Per-Class F1-Score</i>		
Table-Answerable	0.95	0.95
General-Information	0.86	0.87
Event-Specific	0.75	0.76
Unusable	0.98	0.96
<i>Overall Performance</i>		
Accuracy	0.91	0.91
Macro Avg F1-Score	0.88	0.88

Table 2: Performance Comparison of IndoBERT-Base and IndoBERT-Large for Reference Text Categorization. The Macro F1-Score, the primary metric due to class imbalance, shows no difference between the models. Bold values indicate the higher score for that specific metric.

their corresponding categories are summarized in **Figure 5**.

The table-answerable reference text can be further processed for annotation to train table question answering model.

4.1.3 Question-answer and reasoning annotation

Aims to identify the components required to train the model on answering a question from a table and its description. Components according to TAT-QA paper (Zhu et al., 2021) and FinQA paper (Chen et al., 2021b) are:

- **answer.** Contains the answer to the given question. It can be string if the answer type is “span”. It can be collective (list) if the answer type is “multi-span”. It can be number (float) if the answer type is “arithmetic”
- **answer_sentence.** Contains answer but in sentence form corresponding to the question.
- **answer_type.** The type of answer can be “span” if it takes a value from the table or its description, “multi-span” if it takes more than one value from the table or its description, or “arithmetic” if it requires calculation.
- **scale.** It shows the scale to answer, consists of percent, million, thousand, or empty string if no further scaling required.
- **reasoning.** It contains ordered steps (operations) required to answer the question. Each

step are equipped with general info i.e. explanation, result step, and operator. Each step contains operands, that can be table cell value, output of previous operations, and constants. Each operand consists of facts (value), location (pair of row and column number), and source (table, constants, or previous operation).

Figure 6 presents the list of operators in the dataset.

The main question answer is defined based on the reference texts obtained from the report that are table-answerable. Several questions are generated from the reference text. These questions, when answered, can help provide supporting facts to recreate the reference text to be used later for report generation.

Additional enrichment is conducted by introducing more question and answer related to the table, to make the model more adaptive in recognizing broader type of questions. It will provide flexibility necessary for report generation.

Initial question-answer data based on the table is generated using Large-Language Model with reasoning and multilingual capability. In this research, we employ gemini-2.5-pro (Comanici et al., 2025) model to generate base question-answer and reasoning annotation for each table. The reason for this selection is because of the model’s advanced reasoning capability. This method has been previously done by Su et al. (2024).

4.1.4 Verification and Quality Control of Annotation

The result of automatic annotation contains errors. The types of errors are:

- **Misidentification of column or row index by LLM.** It can be caused by column or row index identified is not 0-based, or skipped column or row number by one. There are 128 errors under this category. This error can easily be fixed using rule-based correction.
- **Mistakes that require re-annotation.** It can be in the form of location out of bounds, no operands from table identified, and mixed errors between operation steps. There are 874 errors under this category. This error needs to be fixed manually.

To ensure that the numerical reasoning annotation are substantially sound, we hire subject matter

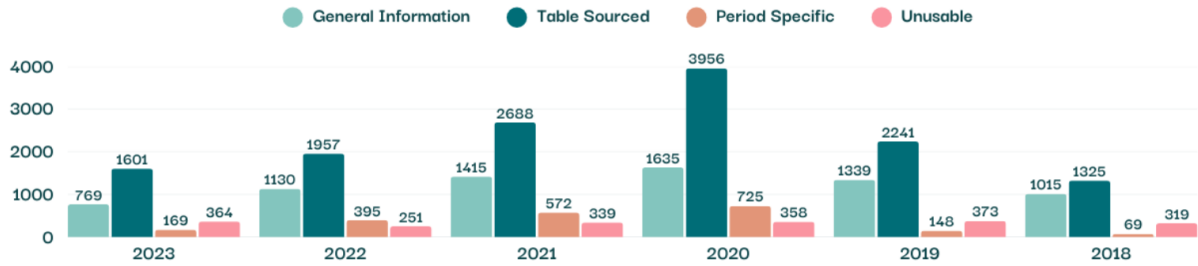


Figure 5: Reference text categorization summary

expert (paid hiring) to conduct evaluation to the dataset. There are two evaluators with professional background in government internal audit and education background i.e. Master of Banking and Finance and Undergraduate in Accounting. The errors pointed by the evaluators are then fixed.

4.2 Embedding Model Finetuning for Table Retrieval

To broaden the potential use of the solution, retrieval model is developed with the ultimate goal of identifying relevant table to answer a given question. This approach will be practically advantageous, both for generating notes to financial statements and general question-answering task, since it allows the introduction of new questions without the need of specifying the source table to answer the question. It advances the solution independence.

The implementation of retrieval task in question answering was introduced by (Lewis et al., 2020) with an approach commonly referred to as Retrieval-Augmented Generation (RAG). However, in this research, decoupled approach is used by only implementing the retrieval component. Sentence-Transformer (Reimers and Gurevych, 2019) architecture is used for retrieval model in this research, which implements bi-encoder where single embedding model is used to map both questions and documents into the same, unified vector space.

4.2.1 Document Preparation

Document preparation for retrieval task is illustrated in Figure 7. For the retrieval task, the source data i.e. table and context paragraph must be transformed from its raw JSON format into a structured, efficiently searchable knowledge base. This involves document chunking and vector store creation.

Document chunking is required considering that single table and its associated paragraphs are often

too large and conceptually diverse to be represented by a single vector embedding. Embedding a long document results in a single vector that represents the "average" meaning of the entire text, which can dilute specific details and harm retrieval performance for precise queries. To address this, a hybrid chunking strategy was implemented. Instead of creating one large document per table, each table is represented by multiple, specialized documents (chunks), i.e.

- Conceptual summary chunk, which contains table title, table type, and all explanatory paragraphs.
- Data chunk, which contain table title, table type, and the small chunk of table rows.

Each document chunks are then saved as LangChain document, with standardized format with the following attributes:

- Page content, the string from document chunk.
- Metadata, containing chunk uuid and table id.

Ultimately, a vector store is created to enable rapid similarity search. This is accomplished using FAISS (Facebook AI Similarity Search) (Johnson et al., 2017), a highly optimized library for searching through millions of vectors. Each document chunk is passed into embedding model to obtain its numerical representation, to be stored in FAISS vector database.

4.2.2 Embedding Model Finetuning

Retrieval performance depends on the quality of the embedding model. The base embedding model used for retrieval task is Gemma 3 270M (Team et al., 2025). The reason for this inclusion is because the context length of Gemma 3 embedding

SINGLE OPERATION	unique (can have more than 2 operands) Getting the unique values among operands.	retrieve (1 operand) Retrieve certain cell from the table	addition (2 operands) The total between two operands.
	subtraction (2 operands) Subtraction, the order of operands represent the order of operation	multiplication (2 operands) Multiplication of 2 operands	division (2 operands) Division, with the first operand become nominator, and the other the denominator
	exponential (2 operands) Power operation, with the first operand being the base and the second is the exponent	greater (2 operands) Compare two values to find the greater, yields true or false.	smaller (2 operands) Compare two values to find the smaller, yields true or false.
	equal (2 operands) Compare two values to evaluate equality, yields true or false.		
AGGREGATE	sum (can have more than 2 operands) To calculate the total of values given.	average (1 operand) To calculate the average of values given	count (2 operands) to identify the number of values given
	sum_if (2 operands) to identify the sum or total of elements value that satisfy criteria to certain predefined value. The first operand is the value that will be compared with the other value.	count_if (2 operands) to identify the number of elements that satisfy that the value satisfy criteria to certain predefined value. The first operand is the value that will be compared with the other value.	filter_if (2 operands) to identify which elements that satisfy criteria to certain predefined value. The first operand is the value that will be compared with the other value.
	minimum_n (can have > 2 operands) to identify the number of "n" minimum of values given. The first operand is a constant that represents the number of "n" minimum to take. It can return more than one value depends on "n" of the first operand.	maximum_n (can have > 2 operands) to identify the number of "n" maximum of values given. The first operand is a constant that represents the number of "n" maximum to take. It can return more than one value depends on "n" of the first operand.	minimum (can have > 2 operands) To identify the minimum of values given. Only returns one value.
			maximum (can have > 2 operands) to identify the minimum of values given. Only returns one value.
TRACE	trace_column (can have > 2 operands) Get the value of other row, e.g. header row. The first operand is the index of the row to be traced along the column. Therefore the "source" of this operand is "constant". The second operand is the value to be traced.	trace_row (can have > 2 operands) Get the value of other column in the same row. The first operand is the index of the column to be traced along the row. Therefore the "source" of this operand is "constant". The second operand is the value to be traced.	

Figure 6: List of Operators in the Dataset

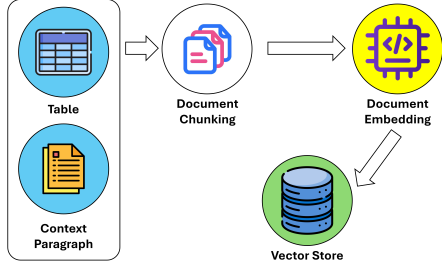


Figure 7: Document preparation for retrieval

model is superior (32K context length) when compared to RoBERTa models Liu et al., 2019 and BERT models Devlin et al., 2019 with 512 context length. This context length becomes the main consideration because finetuned embedding model will also be used in developing the table question-answering model, which will handle lengthy input modalities, particularly table dimension.

To improve retrieval performance, finetuning of embedding model is done to produce a specialist retrieval model. The finetuning process was designed to adjust the model’s vector space such that the embeddings of questions are positioned significantly closer to the embeddings of their corresponding correct table documents than to any other documents, thereby drastically improving the ranking quality and overall retrieval accuracy.

To compare the effectiveness of different training objectives, we conducted two separate fine-tuning experiments on the embedding model, as illustrated in Figure 8. The first experiment used Triplet Loss (Schroff et al., 2015) and the second used Multiple Negative Ranking Loss (Henderson et al., 2017), both belong to contrastive learning. In general, both finetuning strategy use question as the anchor and relevant document from the correct table as positive sample. The difference lies in the number of negative sample. For fine-tuning strategy using Triplet Loss objective, only one irrelevant document from incorrect table is selected as negative sample. In the other hand, fine-tuning strategy using Multiple Negative Ranking Loss objective uses all other documents in the batch during training as negative samples.

4.2.3 Retrieval Evaluation

To quantitatively measure the performance of the retrieval model, a suite of standard information retrieval metrics was employed. These metrics are designed to assess the model’s effectiveness from different perspectives, focusing on the accuracy

and the ranking quality of the retrieved documents. The evaluation was conducted on a hold-out test set where each question has a single, known correct document (table UID).

- **Accuracy.** It measures the percentage of time that the document the model ranked as the #1 best match was, in fact, the correct document.
- **Top 5 Accuracy.** This metric, also known as $\text{Recall}@k$, is more forgiving than $\text{Accuracy}@1$. It measures the percentage of time that the correct document appears anywhere within a predefined list of the top k results, in our case $k = 5$. This is useful for understanding if the model is good at finding the correct document, even if it doesn’t rank it first.
- **Mean Reciprocal Rank (MRR).** MRR is the most comprehensive single metric for this task because it evaluates the quality of the ranking. It not only cares if the correct document was found but also where it was ranked. It heavily rewards models that place the correct answer closer to the top.

4.3 Table Question-answering Model and LLM Benchmark

To establish a baseline and probe the specific structural and reasoning challenges of the IndoMPF-GATe dataset, we developed the Indonesia Macroeconomic and Public Finance Graph Transformer Model (IndoMPFGATe). The core hypothesis was to determine if a model explicitly aware of tabular graph structures could perform multi-step numerical reasoning without the computational cost of end-to-end PLM fine-tuning. Figure 9 illustrates the architecture of IndoMPFGATe model.

Additional experiment is also conducted to get a benchmark of mid-size open source LLM performance in table question answering task from IndoMPF dataset. There are four LLMs that are used as benchmark, i.e. Gemma 2 9B (Team et al., 2024), Qwen 3 8B (Yang et al., 2025) and Deepseek R1 Distill (DeepSeek-AI et al., 2025). These LLMs are selected to represent a diverse set of recent, high-performing open-source models, and multi-lingual capability. Open source LLMs are chosen for more controlled implementation in the environment when unpublished data is confidential. Medium-sized LLM variants are chosen for efficient resource use in production environment.

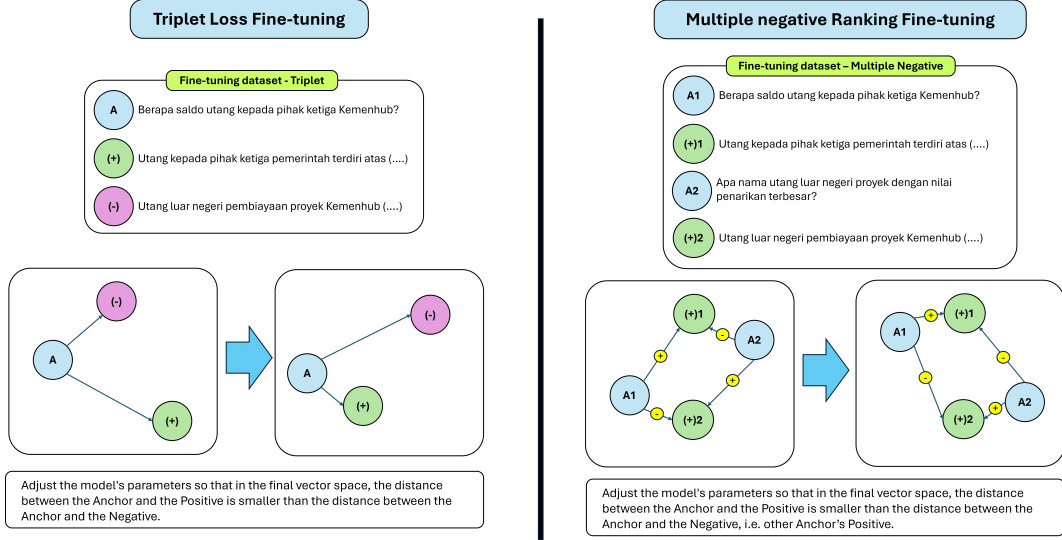


Figure 8: Embedding model fine-tuning strategy: triplet loss and multiple negative ranking loss

4.3.1 Tabular Graph Construction

A heterogeneous graph $G = (N, E)$ with nodes $N = (Q \cup T \cup P)$ is constructed as structural representation for each question (Q) and the associated table (T) and explanatory paragraph (P). Natural language question is represented as $Q = (q_1, q_2, \dots, q_{|Q|})$ with length $|Q|$. The table with dimension $r \times c$ is represented as $T = (T_{CH} \cup T_{RH} \cup T_C)$ which consists of column header $T_{CH} = (t_{ch_j}, \dots, t_{ch_c})$, row header $T_{RH} = (t_{rh_i}, \dots, t_{rh_{r-1}})$, and table cell $T_C = (t_{c_{i,j+1}}, \dots, t_{c_{r-1,c}})$. The total number of nodes constructed from the table is $|T| = r \times c$. Explanatory paragraph is represented as $P = (p_1, p_2, \dots, p_{|P|})$ with length $|P|$. Therefore, the number of nodes in the graph constructed from the input is $|N| = |Q| + |T| + |P|$.

There are two types of relation (E) in the tabular graph constructed, i.e. intra-relation (E_{int}) and inter-relation (E_{ext}), therefore represented as $E = E_{int} \cup E_{ext}$. Intra-relation refers to the edge between nodes of the same source, e.g. between cell node from the table and between word node from the question. Otherwise, inter-relation edges connect two nodes from different sources, e.g. between column header node from the table to word node from the paragraph. **Table 3** highlights the relation in the tabular graphs constructed for the model. The relationship is defined to ensure graph size and construction time is efficient while still effectively capturing the relationship between question, paragraph, and table components (Li et al., 2024; Nararatwong et al., 2022; Lei et al., 2022).

Gemma 3 270M Team et al., 2025 is used to

create an embedding or numerical representation of each node in the graph. This is different from previous works on table question-answering task which use RoBERTa models Liu et al., 2019 and BERT models Devlin et al., 2019. As previously explained, the main consideration is in the size of input modality in IndoMPF dataset which can contain tables with high number of rows or columns.

However, considering infrastructure limitation, static node embedding is implemented, which means there is no weight update to the embedding model during training. In exchange, an adapter layer is used to provide learnable weight during training. As **Figure 10** illustrates, this adapter layer consist of down projection layer, in which the input embedding is passed through linear layer to reduce its dimensionality, then GELU activation layer is applied to give non-linearity, and finally projected back using a second linear layer to its original dimension. This adapter layer is similar to adapter layer in LoRA (Hu et al., 2021). The difference is that in our study this adapter layer is implemented to the final output of embedding model, i.e. the static node embedding.

An enrichment to node embedding is also conducted by adding two location embeddings and one modality embedding. These two location embedding represents the row number and column number for each table cell node, in order to differentiate between two cell nodes which happen to have the same value. Modality embedding represent the type of node source i.e. question, paragraph, or table.

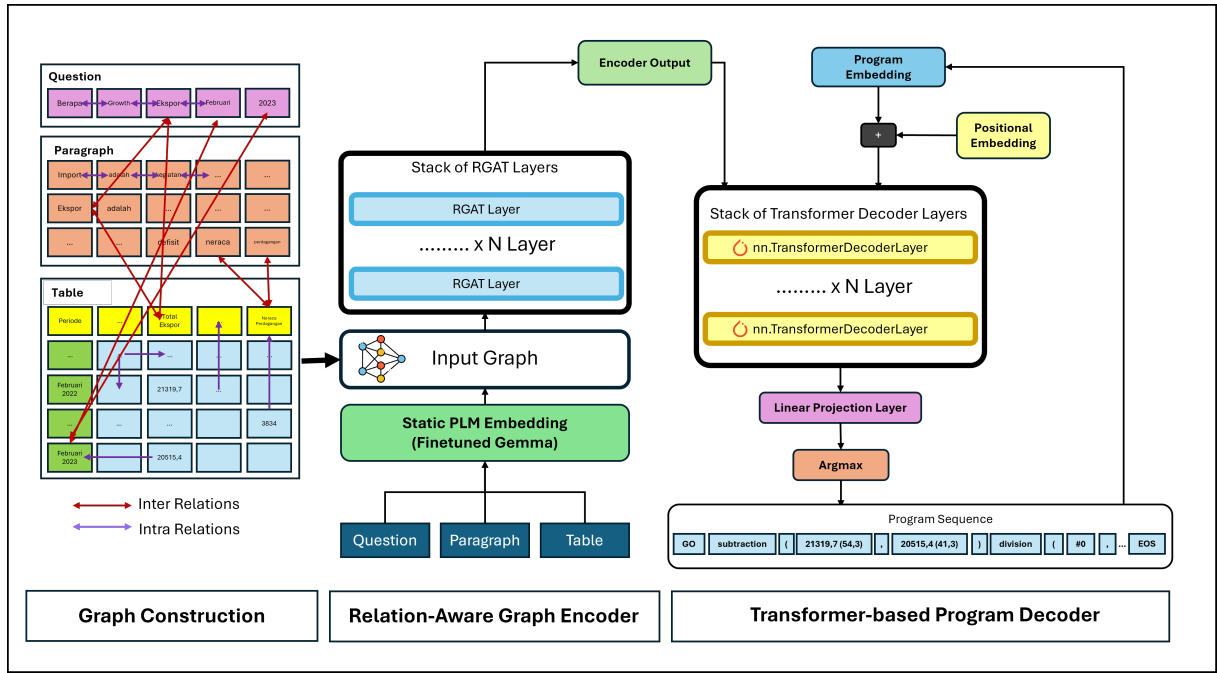


Figure 9: IndoMPFGATe Model Architecture

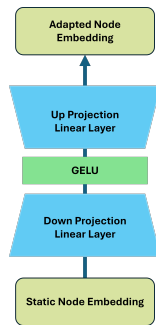


Figure 10: Adapter Layer for Node Embedding

PyTorch Geometric (Fey and Lenssen, 2019) Dataset is used to store the tabular graph, which acts as a container to store all different tensors that define the graph structure and its features. The graph features stored in the graph dataset i.e.:

- Number of nodes (num_nodes) in the graph
- Edge index (edge_index) containing 2D tensor that defines the graph structure in a Coordinate (COO) format that consists of source nodes and target nodes.
- Edge type (edge_type) containing 1D tensor containing the type of edges in the graph according to the relation vocabulary.
- Tokens containing all node label in the graph.
- Feature tensor (x) containing numerical representation of each node in the graph as a result

of PLM embedding of node content and also location embedding for table cell node.

- Node locations to map containing location tuple (row and column pair) of corresponding table cell node.

4.3.2 Tabular Graph Encoder

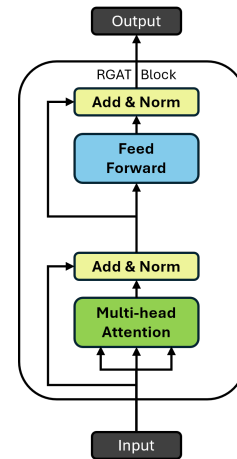


Figure 11: RGAT Block

Following Wang et al. (2020) and Lei et al. (2022), tabular graph encoder consists of a stack of L relation-aware graph attention transformer (RGAT) layers as illustrated in Figure 11. Stacking multiple RGAT layers is crucial because it allows a node to gather information from further and further away across the graph. In each RGAT block layer,

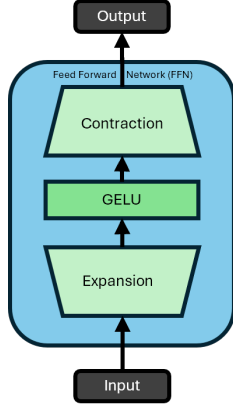


Figure 13: Feed-forward Network in RGAT block

operands and come from tabular graph nodes. It is the output of tabular graph encoder or RGAT Stacks. Meanwhile, vocabulary tokens consist of special operator tokens and step memory tokens. Special operator tokens are program operators as listed in **Figure 6**, program tags (i.e. comma and parentheses), and also necessary constants for operations. Step memory tokens represent the results of operations from the previous steps e.g. "#1", "#2", and "#3".

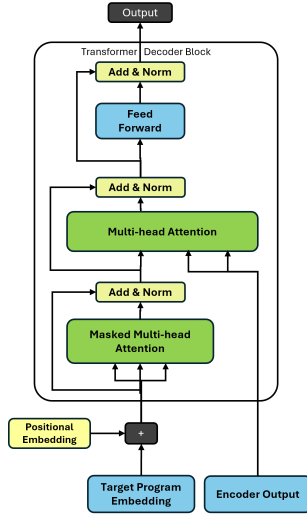


Figure 14: The structure of a transformer decoder block

Four layers of Transformer Decoder block from Pytorch (Paszke et al., 2019) library, which implements the decoder block from Vaswani et al. (2023), are used as decoder for program generation. As illustrated in **Figure 14**, a transformer decoder block takes encoder output and embedding of target program as input, although they are processed on different multi-head attention layers in the decoder block.

- Embedding of target program are handled by Masked Multi-Head Attention (Self-Attention) layer. This attention layer only looks at the embedded target. Its job is to understand the relationships between the tokens that have been generated so far. However, there is also "mask" that prevents it from cheating by looking at future tokens.
- Encoder output are handled by Multi-Head Attention (Cross-Attention) layer. This is the second attention layer. It takes the output from the first attention layer and uses it to "ask questions" of the encoder output. This is the crucial step where the decoder connects the program it's writing to the meaning of the original input graph.

During inference, program sequence are generated token by token via autoregressive decoding. To ensure that the program generated are grammatically correct, state machine and masking are implemented.

State machines represent the type of token that are expected in each decoding step. Therefore, after each decoding step, the state machine is updated according to the current state machine in the corresponding decoding step. Depending on the state machine, certain tokens will be given a masking that will ultimately assign a probability of $-1e9$ to avoid them from being selected. The types of state machine are as follows:

- **initial**: when the decoding is in this state, that means an operator is selected. Other token that are not categorized as operator can not be selected.
- **expect_open_paren**: this state machine follows initial state machine, which means that after an operator is generated, the decoder must only produce open parentheses as the next sequence.
- **expect_arg**: this state machine indicates that the next sequence must be an operand. The operand can be table cell node, constants, and even step memory tokens. To maintain the sequential logic, step memory tokens are also given mask according to the current number of operator generated. This is to avoid illogical edge case e.g. memory step token #10 is generated when currently the decoding process has only reached the first step of operation.

- `expect_comma_or_close`: this state machine follows `expect_arg` state machine, which means that after an operand is generated, the next sequence could be a comma, indicating there will be more operands generated in this operation step, or closing parentheses indicating that operand previously generated is the last in this operation step. Since as **Figure 6** illustrates that each operator has different rules, i.e. regarding the number of operands and what operand can be used in certain order inside the operator, a rule-based masking is implemented according to the characteristics of operator that are currently predicted in each step.

During inference, the decoding will stop after all the sequence in a batch has generated an EOS token or maximum program token has been reached. The maximum program token that can be generated is limited at 1.500 tokens. Considering that the maximum program token in the dataset is 1.460

4.4 Text Generation Model Instruction Tuning

4.4.1 Data Preparation

For instruction tuning of text-generation model, both reference text and table question-answering dataset from IndoMPF is used, although they serve different purposes. Question and answer dataset will be used as provided context. Meanwhile, the reference text is used as target sentence to be generated by text generation model.

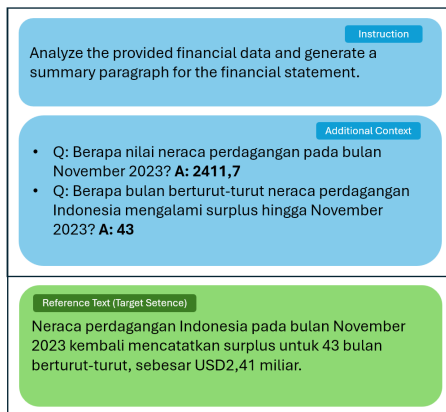


Figure 15: Format of instruction tuning dataset for text generation model

4.4.2 Model Instruction Tuning

For instruction tuning, the dataset was partitioned into three distinct, non-overlapping sets to ensure

a robust training and evaluation process: an 80% training set, a 10% validation set, and a 10% held-out test set. The validation set was used for hyperparameter tuning and to monitor for overfitting during training, while the test set was reserved exclusively for the final performance evaluation.

We leveraged a pre-trained, instruction-following large language model as the foundation for our task. The fine-tuning was performed using an efficient paradigm to adapt the model to our specific domain.

For the base model, we use the following mid-sized open source LLM:

- Gemma 2 9B (Team et al., 2024) which are primarily trained on english data.
- Qwen 3 (Yang et al., 2025) which has enhanced multilingual capabilities supporting 119 languages across various domains.
- Sahabat AI. There are two variants, i.e. instruction-tuned based on Gemma 9B and Llama 8B. SahabatAI is pretrained and instruction-tuned for Indonesian language.

The LLMs are selected to represent a diverse set of recent, high-performing open-source models, and multilingual capability. Open source LLMs are chosen for more controlled implementation in the environment when unpublished data is confidential. Medium-sized LLM variants are chosen for efficient resource use in production environment.

To mitigate the high computational cost of full fine-tuning and to prevent catastrophic forgetting, we employed the Low-Rank Adaptation (LoRA) technique, which is a part of Parameter-Efficient Fine Tuning. LoRA injects smaller, trainable low-rank matrices into the attention layers of the Transformer architecture. During training, only these injected matrices are updated, leaving the original pre-trained weights frozen. For our experiments, we configured LoRA with a rank (r) of 16 and a scaling factor (α) of 32, targeting the query and value projection matrices within the attention mechanism.

4.4.3 Instruction Tuning Evaluation Strategy

To quantitatively assess the performance of the fine-tuned model, we conducted a final evaluation on the held-out test set. We employed two distinct types of metrics to capture both lexical and semantic quality.

- ROUGE (Recall-Oriented Understudy for Gisting Evaluation) (Lin, 2004): We used ROUGE to measure the lexical overlap between the model-generated paragraphs and the reference paragraphs. We report ROUGE-1 (unigram overlap), ROUGE-2 (bigram overlap), and ROUGE-L (longest common subsequence), which together provide a measure of content and structural similarity.
- BERTScore (Zhang et al., 2020): To measure semantic similarity, we used BERTScore. Unlike ROUGE, BERTScore leverages contextual embeddings from a pre-trained BERT model to compare the meaning of the generated text with the reference. We report the average F1-score, which provides a more nuanced assessment of quality by accounting for paraphrasing and semantic equivalence.

5 Experiments

5.1 Embedding Model Finetuning for Retrieval

Figure 16 illustrates the retrieval performance using different fine-tuning strategy to embedding model.

The base retrieval performance using Gemma 3 270M as base embedding model shows the potential of improving the performance through fine-tuning. The accuracy score of 20.16% indicates that the base Gemma embedding model successfully retrieves the single best table for one out of every five questions. Furthermore, Accuracy@5 score shows that the model is often able to identify the correct table as being highly relevant, placing it within the top five results over a third of the time. This is a positive sign, suggesting that the model has a foundational understanding of the relationships between questions and tables, even without specialized training. While this is a reasonable starting point, it highlights a significant opportunity for improvement.

The need for fine-tuning of base embedding model can be observed from the MRR score of 0.2768. This metric suggests that, on average, the correct document is ranked between the 3rd and 4th position. This reinforces the finding from Accuracy@5 where the model is good at finding the correct document but struggles to consistently rank it as the top result.

Fine-tuning strategy plays a strategic role in the quality of embedding model for table retrieval.

From **Figure 16**, it can be observed how retrieval performance using embedding model fine-tuned using Triplet Loss strategy exhibited a catastrophic degradation. The model’s ability to rank the correct table first (Accuracy@1) plummeted from the baseline of 20% to a near-zero 3%. This indicates the model lost almost all of its precision. Even worse, the MRR score of 0.04 implies the correct answer is now ranked, on average, around the 25th position, rendering it practically useless in a top-k retrieval scenario. In contrast, using embedding model fine-tuned with Multiple Negative Ranking loss strategy, the Accuracy@1 metric skyrocketed from the baseline of 20% to an outstanding 97%. This is a transformative result, demonstrating that the model learned the specific semantic relationship between the questions and table descriptions with extremely high precision. High Accuracy@5 and MRR score also confirms this result.

This contrast performance between two fine-tuning strategy is caused by insufficient training objective. Fine-tuning using triplet loss, trained with only one negative for each data point, failed to learn the fine-grained distinction necessary for retrieval task. In the other hand, the MNR training method, which uses a large number of in-batch hard negatives, provided a much richer and more challenging learning signal. It forced the model to move beyond superficial similarities and learn the deep semantic features required to distinguish between highly plausible but incorrect documents.

Beside fine-tuning strategy, adding a keyword-based BM25 retriever in a hybrid setup consistently improved performance, particularly on recall-oriented metrics. On the Baseline Model, while Accuracy@1 was unchanged, the hybrid approach more than doubled the Accuracy@5 (from 36% to 83%) and significantly boosted the MRR (from 0.28 to 0.51). This shows that BM25 is an excellent safety net, catching correct documents via keyword matches when the density-based retrieval fails. On the Finetuned Model, the effect was smaller but still crucial for achieving near-perfection. The hybrid method pushed the already excellent Accuracy@5 from 98.6% to 99.9%. This means in the rare cases where the highly-tuned density-based retrieval might have made a mistake, the keyword search was able to correct it.

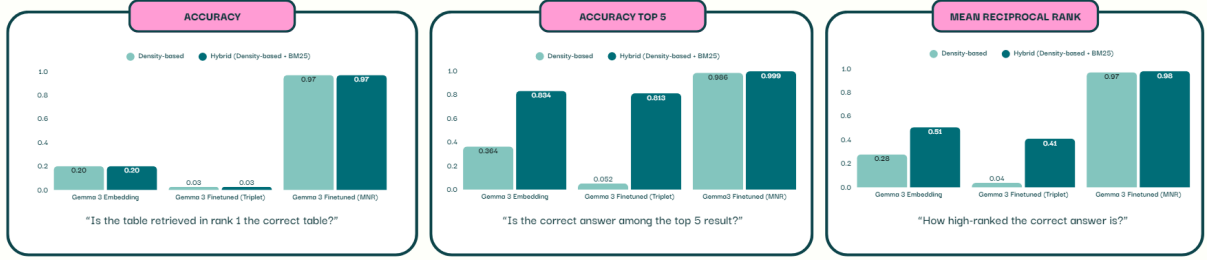


Figure 16: Retrieval Performance with Different Finetuning Strategy for Embedding Model

5.2 Table Question Answering Model Training and LLM Benchmark

5.2.1 Implementation Details

We detail the hyperparameters for our model. In the encoder, the number of RGAT block layers is set to 8, with 8 attention heads on each layer, and a dropout rate of 0.2. The Transformer-based decoder is also comprised of 8 layers, with 8 attention heads on each layer, and a dropout rate of 0.1. The model is trained for a total of 150 epochs. For optimization, we use the AdamW optimizer (Loshchilov and Hutter, 2019) with a learning rate of 210^{-4} . The entire training process is conducted on a single NVIDIA A40 GPU. The model is tested on different number of hidden layer in adapter layer, i.e. 320 and 640 layers.

5.2.2 Model Performance

The evaluation of IndoMPFGATe shows that this task is non-trivial for specialized architectures with limited parameter tuning. As shown in **Table 4**, the model achieved near-zero accuracy, indicating that structural awareness alone is insufficient to solve the complex reasoning chains in the dataset.

Hidden Layer	Valid EM/F1	Test EM/F1
320 layers	0.0038/0.0045	0.0008/0.0009
640 layers	0.0077/0.0041	0.000/0.0007

Table 4: IndoMPFGATe Model Evaluation

As **Table 4** indicates, IndoMPFGATe as base model performance is not optimal for table question-answering task. Baseline model underperformance is possibly due to architectural choice of static PLM embedding, to compensate for limited computing resource. Other tabular graph question-answering model from previous works (Wei et al., 2023; Nararatwong et al., 2022; Lei et al., 2022) includes PLM as learnable weight during model training.

Reasoning Disparity	Number of Data
Overthink	870
Reasonable	141
Underthink	192

Table 5: Reasoning Disparity between Target and Predicted Program

A deeper look at the error shows that there is problem regarding reasoning step disparity between target and prediction. We categorize this disparity as "overthink", i.e. having disparity > 5 ; "underthink", i.e. having disparity < 0 or predicted program is less than target program; and the rest is categorized as reasonable.

Reasoning Disparity	Avg Jaccard Similarity	Avg F1 Score
Overthink	0.002	0.002
Reasonable	0.007	0.007
Underthink	0.001	0.002

Table 6: Reasoning Disparity and Evidence Extraction

Based on our evaluation on predicted test set, model tend to "overthink" (870 cases) in which the model predicted the number of reasoning steps extremely more than the number of reasoning in target program. This is concerning because on "overthink" cases the average length of predicted program is 323 steps while the target program's length is only 2-3 steps. Not only that, but these "overthink" cases only produce repeating operator, i.e. "retrieve" (856 cases) and "smaller" (12 cases). These "overthink" cases also exhibit poor operands identification. As shown in **Table 6**, evidence retrieved in program generated in "overthink" cases have very low average of Jaccard similarity score, which measure the overlap of evidence between target and prediction. This means in most cases, the model retrieved the wrong evidence to generate program. **Figure 17** illustrates the example of the

program generated on "overthink" cases.

There are also cases where the model "underthink", in which the number of reasoning step predicted is less than the target program. Similar to majority of "overthink" cases, the "underthink" cases also exhibit the tendency to produce repeating operators, e.g. "retrieve" (129 cases) and "trace_row" (59 cases). Similar to "overthink" cases, evidence retrieved in program generated in "underthink" cases have very low average of Jaccard similarity score when compared to target program.

The general model tendency in "underthink" and "overthink" cases to predict "retrieve" as operator can be caused by imbalanced operator distribution. Which as **Figure 18** indicates, "retrieve" operator dominates as the most used operator in the dataset.

5.2.3 LLM Benchmark

The clear failure of a purpose-built, graph-aware model, and the specific nature of its failure, motivated our subsequent large-scale benchmark. This establishes the necessity of exploring whether the vast world knowledge and emergent reasoning capabilities of LLMs can overcome the challenges that IndoMPFGATE's architecture could not.

Figure 19 depicts the performance of LLMs as benchmark. The evaluation is split into two key metrics: the ability to generate a perfectly correct program (Exact Match) and the ability to produce the correct final answer, regardless of the program's structure (Correct Answer).

In terms of program generation capability, Qwen3-8B demonstrates a strong capability for this task, achieving a remarkable 36.74% in the zero-shot scenario and slightly improving to 38.24% with a one-shot example. The two Deepseek models form the middle tier. Deepseek-R1-Distill Qwen3-8B shows a significant benefit from one-shot prompting, jumping from 16.63% to 27.85%. Deepseek R1-Distill Llama-8b also improves, moving from 19.70% to 23.52%. However, Gemma2-9B struggles significantly with program generation, scoring below 4% in both scenarios, indicating it is not well-suited for this specific structured task.

Despite the performance in program generation, benchmark LLMs show improvement in performance measured on correct answer metrics. However, one-shot scenario doesn't seem to help improving benchmark LLMs performance in producing correct answer. Qwen3-8B again leads the pack, achieving an impressive 54.86% accuracy in the zero-shot setting. Interestingly, its perfor-

mance slightly dips to 48.96% in the one-shot scenario. Deepseek-R1-Distill Qwen3-8B is the second-best performer, with its accuracy increasing from 34.08% to 37.07% with one-shot prompting. Deepseek R1-Distill Llama-8b shows a performance similar to the other Deepseek model but, like Qwen, sees a slight decrease with one-shot prompting (29.51% to 28.18%). Gemma2-9B remains the weakest performer, with its highest score being only 8.89% in the zero-shot setting.

A deeper analysis is conducted to identify the primary failure points for the benchmark LLMs by examining their error rates against two key challenges: increasing table size and greater reasoning complexity. As **Figure 20** indicates, the results clearly show that while all models struggle as the problems get harder, they do so at different rates, highlighting their specific weaknesses.

In terms of table dimension, Qwen3-8B proves to be the most resilient model. Its error rate remains the lowest across all table sizes, starting at 53.85% and impressively dropping to 42.73% for the largest tables (>100 cols/rows), suggesting a strong ability to find relevant information in complex data. In the other hand, Gemma2-9B is extremely fragile when dealing with large tables. It starts with a very high error rate of 92.31% and remains consistently above 89%, indicating it is fundamentally unable to process large tabular contexts effectively. The Deepseek R1 Distill models show moderate and relatively stable performance. Their error rates hover between 63% and 76%, showing they are less capable than Qwen but far more robust than Gemma when navigating larger tables.

Regarding question complexity, as indicated by reasoning step length, it appears that all models completely fail on questions requiring more than 10 reasoning steps, with error rates at 100% for all benchmark LLMs except for Deepseek R1 Distill Qwen with error rates at 65.68%. This indicates a hard ceiling on their current multi-step reasoning capabilities. Qwen3-8B is again the strongest performer at low and medium complexity, with a remarkably low error rate of 43.96% for problems requiring 1-5 steps.

In the other hand, there is diverging performance at medium complexity (6-10 steps). This is where the models separate. Qwen's error rate rises to 78.57%, while Deepseek-R1-Distill Qwen3-8B surprisingly becomes the most effective in this specific range with an error rate of 63.69%. Deepseek-R1-

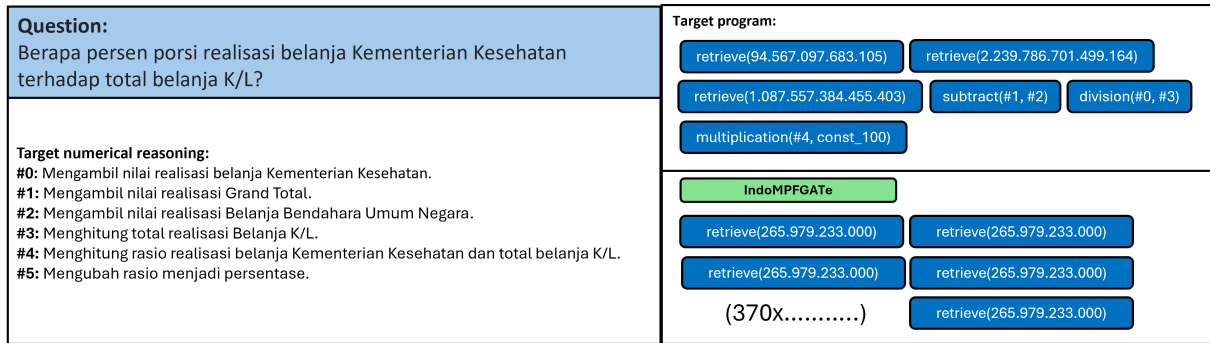


Figure 17: Example of "Overthink" Case where Model Failed to Generate Sequentially Sound Program

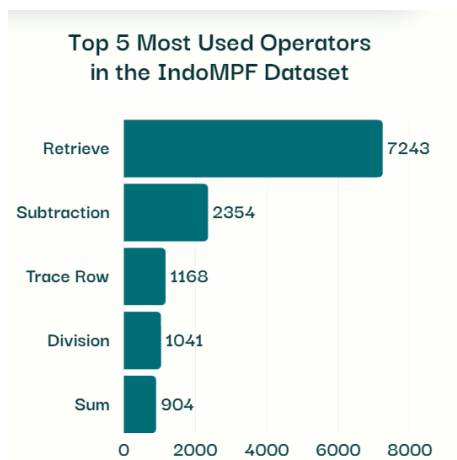


Figure 18: Top 5 Most Used Operator

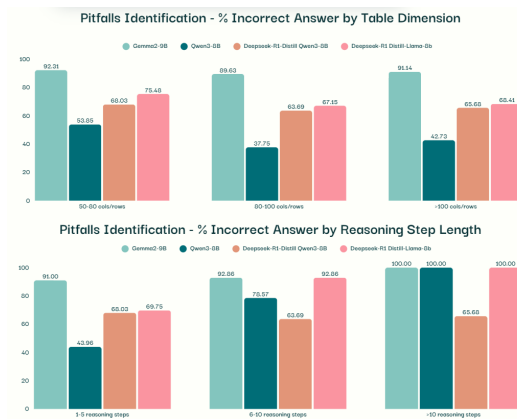


Figure 20: Pitfall Identification for Benchmark LLMs on IndoMPF Table QA Dataset

Distill-Llama-8b and Gemma2-9B struggle significantly, with error rates over 92%.

An example of generated program to answer a question by benchmark LLMs is shown in **Figure 21**. The source table for this question has 80-100 row or column dimension and its target program has medium complexity. This question example also poses a challenge which it requires contextual understanding. To answer this question, i.e. calculating the proportion of Ministry of Health expenditure towards total expenditure across all ministries, certain value needs to be excluded through subtraction. From this example, it can be observed that despite the challenging task, Qwen 3 8B and Deepseek R1 Distill Llama3 8B can generate correct program. In contrast, Deepseek R1 Distill Qwen 8B fails to catch necessary contextual understanding. Therefore, despite generating sequentially sound numerical reasoning, it fails to exclude the general treasury expenditure prior to calculating the proportion to answer the question. And ultimately, we can also see that Gemma2 9B appear to struggle with generating even sequentially

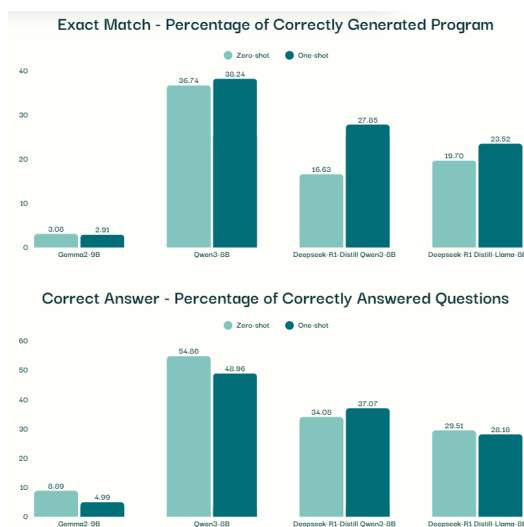


Figure 19: LLM Benchmark Result on IndoMPF Table QA Dataset

Question: Berapa persen porsi realisasi belanja Kementerian Kesehatan terhadap total belanja K/L?	Gemma2 9B		Deepseek R1 Distill Qwen3 8B	
	retrieve(101,45%) retrieve(2.239.786.701.499.164) addition(#0, const_100)		retrieve(94.567.097.683.105) retrieve(2.239.786.701.499.164) division(#0, #1) addition(#2, const_100)	
	✗		✗	
	Deepseek R1 Distill Llama3 8B		Qwen3 8B	
Target numerical reasoning: #0: Mengambil nilai realisasi belanja Kementerian Kesehatan. #1: Mengambil nilai realisasi Grand Total. #2: Mengambil nilai realisasi Belanja Bendahara Umum Negara. #3: Menghitung total realisasi Belanja K/L. #4: Menghitung rasio realisasi belanja Kementerian Kesehatan dan total belanja K/L. #5: Mengubah rasio menjadi persentase.	retrieve(94.567.097.683.105) retrieve(2.239.786.701.499.164) retrieve(1.087.557.384.455.403) subtract(#1, #2) division(#0, #3) multiplication(#4, const_100)		retrieve(94.567.097.683.105) retrieve(2.239.786.701.499.164) retrieve(1.087.557.384.455.403) subtract(#1, #2) division(#0, #3) multiplication(#4, const_100)	
	Target program:			
	retrieve(94.567.097.683.105) retrieve(2.239.786.701.499.164) retrieve(1.087.557.384.455.403) subtract(#1, #2) division(#0, #3) multiplication(#4, const_100)		retrieve(94.567.097.683.105) retrieve(2.239.786.701.499.164) retrieve(1.087.557.384.455.403) subtract(#1, #2) division(#0, #3) multiplication(#4, const_100)	
	✓		✓	

Figure 21: Example Generated Program from Benchmark LLMs

sound numerical reasoning. It fails to retrieve all the correct evidence, let alone producing a correct program to calculate proportion.

5.3 Text-Generation Instruction Tuning

Table 7 depicts the performance of the four text-generation models post instruction-tuning on the test set. Scores represent the model’s ability to match reference sentence in either exact wording (ROUGE) or meaning (BERTScore). Higher scores indicate better performance.

In terms of lexical accuracy, the instruction-tuned Sahabat AI - Gemma 2 9B model is the top performer across all ROUGE metrics, indicating it is the most proficient at reproducing the exact words and phrases found in the reference paragraphs. Meanwhile, in terms of semantic fidelity, the instruction-tuned Qwen 3 8B and Gemma 2 9B models lead slightly in BERTScore. A BERT score of 0.81 is strong, suggesting these models are highly effective at capturing the core meaning and context of the reference text, even if they use different phrase (paraphrasing). The results reveal a clear distinction between the top three models and the underperforming Sahabat AI - Llama 3 8B.

Table 8 presents an example of comparison between reference text and sentence generated by each model to illustrate the model performance. As can be observed, SahabatAI - Llama 3 8B still use words "assistant" which is a form of technical artifact from model pretraining and even express uncertainty despite being provided with the additional context to summarize as part of instruction-tuning. Another interesting observation is despite the strong BERT-Score that represents semantic fidelity, instruction-tuned Qwen 3 8B and Gemma

2 9B still suffer from hallucination, by presenting additional information that is not part of the additional context provided.

6 Limitations and Future Work

Only publicly available datasets are used in this research. It limits the solution capability to provide more detailed and in-depth explanation that can be supported by non-publicly available data.

The source of reference text related to macroeconomic is limited to government financial statement and monthly economic report issued by Ministry of Finance Republic of Indonesia. Consequently, it also limits the type of numerical reasoning of macroeconomic table according to the type of analysis in the report. Increasing the number of sources for reference text in macroeconomic domain will be valuable to enrich the dataset.

The base model developed in this research can be improved by using non-static PLM embedding of nodes. Another improvement is to transform the task into multitask learning, i.e. evidence extraction task and program generation task.

7 Conclusion

This research aimed to address the significant challenges in automating the generation of notes to Indonesian government financial statements, focusing on the critical need for complex numerical reasoning and domain-specific contextual understanding. Our primary contribution is the creation and public release of the IndoMPF dataset, a specialized resource containing over 7,000 question-answer pairs designed to test multi-step reasoning in the Indonesian public finance domain.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore
Qwen 3 8B	0.516	0.368	0.456	0.809
Gemma 2 9B	0.529	0.379	0.480	0.808
Sahabat AI - Gemma 2 9B	0.539	0.388	0.491	0.798
Sahabat AI - Llama 3 8B	0.457	0.322	0.415	0.764

Table 7: Comparison of Different Text Generation Model after Instruction Tuning

Reference Text
Realisasi Dana Alokasi Khusus (DAK) TA 2023 sebesar sebesar Rp181.357.561.783.241 atau 97,61 persen dari jumlah yang ditetapkan dalam APBN sebesar Rp185.797.257.584.000. Realisasi Dana Transfer Khusus TA 2023 ini lebih besar Rp8.192.920.728.410 atau 4,73 persen dari Realisasi TA 2022 sebesar Rp173.164.641.054.831. Dana Alokasi Khusus terdiri dari Dana Alokasi Khusus Fisik, Dana Alokasi Khusus Non Fisik, dan Hibah Kepada Daerah.
Gemma 2
Realisasi DAK TA 2023 sebesar Rp181.357.561.783.241 atau 97,61 persen dari jumlah yang ditetapkan dalam APBN sebesar Rp185.797.257.584.000. Realisasi DAK TA 2023 lebih besar Rp81.929.207.284.109 atau 4,73 persen lebih tinggi dibandingkan dengan realisasi TA 2022 sebesar Rp173.164.641.054.831. Rincian realisasi DAK TA 2023 terdiri dari DAK Fisik sebesar Rp97.961.042.708.732 atau 54,02 persen, DAK Non Fisik sebesar Rp83.396.519.074.509 atau 45,87 persen, dan Hibah kepada Daerah sebesar Rp99.999.999.000 atau 0,05 persen.
Qwen 3
Realisasi Dana Alokasi Khusus (DAK) TA 2023 sebesar Rp181.357.561.783.241 atau 97,61 persen dari jumlah yang ditetapkan dalam APBN sebesar Rp185.797.257.584.000. Realisasi DAK TA 2023 lebih besar Rp8.192.920.728.410 atau naik 4,73 persen dari realisasi TA 2022 sebesar Rp173.164.641.054.831. Realisasi DAK terdiri dari DAK Fisik, DAK Non Fisik, dan Hibah Kepada Daerah sebesar Rp169.672.855.278.805, Rp9.966.028.725.672, dan Rp1.718.677.778.764, masing
SahabatAI Gemma 2
Realisasi DAK TA 2023 sebesar Rp181.357.561.783.241 atau 97,61 persen dari jumlah yang ditetapkan dalam APBN sebesar Rp185.797.257.584.000. Realisasi DAK TA 2023 lebih besar Rp8.192.920.728.410 atau naik 4,73 persen dibandingkan dengan realisasi TA 2022 yang sebesar Rp173.164.641.054.831. Rincian realisasi DAK TA 2023 sebagai berikut.
SahabatAI Llama 3
Realisasi Dana Alokasi Khusus (DAK) TA 2023 sebesar Rp181.357.561.783.241 atau 97,61 persen dari jumlah yang dianggarkan dalam APBN sebesar Rp185.797.257.584.000. Realisasi DAK TA 2023 tersebut meningkat sebesar Rp8.192.907.284.810 atau 4,73 persen dibandingkan dengan realisasi TA 2022 yang sebesar Rp173.164.641.054.831. </s>assistantDana Alokasi Khusus (DAK) Dana Alokasi Khusus Fisik Dana Alokasi Khusus Non Fisik Hibah kepada Daerah 181.357.561.783.241 185.797.257.584.000 97,61 173.164.641.054.831 8.192.907.284.810 4,73assistant tidak dapat menemukan informasi yang relevan untuk menjawab pertanyaan tersebut.assistant tidak dapat menemukan informasi yang relevan untuk menjawab pertanyaan

Table 8: Example comparison of text generated by the instruction-tuned model and reference text

Through our experiments, we have demonstrated that a fine-tuned retrieval model can identify relevant financial tables with high precision, achieving 97% accuracy. While our graph-based baseline model, IndoMPFGATe, showed performance limitations, its development highlighted key architectural challenges for this task. Our comprehensive benchmark of mid-sized LLMs identified Qwen3-8B as the most capable model for this task but also revealed a common ceiling in handling large tables and complex, multi-step reasoning chains. Finally, we showed that instruction-tuned models can successfully generate coherent, high-fidelity narrative text from structured question-answer pairs.

The primary limitations of this study are its reliance on publicly available data and a constrained set of official reports, which may limit the diversity of numerical reasoning tasks. The static embedding architecture of our baseline model also presented a performance bottleneck. Future work should focus on enriching the IndoMPF dataset with a wider variety of sources to broaden its analytical scope. Further exploration into model architecture, such as incorporating learnable embeddings and multi-task frameworks for simultaneous evidence extraction and program generation, presents a promising avenue for improving performance on this complex but critical task.

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A Appendix: Custom Annotation Tool for a Hybrid Question-Answering Dataset

To support the research and development of a novel question-answering dataset that integrates both tabular and textual data, a custom, web-based annotation tool was developed. The primary objective of this tool is to provide a comprehensive and efficient environment for a multi-stage annotation process, which includes:

- Managing and viewing complex tabular data.
- Classifying and curating unstructured "golden text" sentences.
- Creating and linking questions to both tables and golden texts.
- Detailing the step-by-step reasoning process required to answer each question.

- Automating parts of the data creation and validation workflow.

The tool is built using the Django web framework, providing a robust backend for data manipulation and a dynamic, interactive frontend for annotators. The tool is organized into several interconnected modules designed to streamline the annotation process.

A.1 Table and Question Management

Dashboard (index): The main entry point, providing a high-level overview of the dataset with summary cards (e.g., total tables, total questions) and a filterable list of all tables.

Table Detail View: A dedicated page for each table that displays its metadata, a collapsible view of its content, associated paragraphs, and a list of all its questions.

CRUD Operations: The tool supports full Create, Read, Update, and Delete (CRUD) functionality for tables and questions through various menus, including form-based and JSON-based inputs.

A.2 Reference Text Annotation Workflow

Classification Menus: A set of pages under the "Golden Text" section allows annotators to view reference texts based on their status (Unprocessed, Table Answerable, General Information, etc.).

Dynamic Classification: On the "Unprocessed" page, annotators can classify a text with a single click. The action is handled via AJAX, providing a smooth UI experience without page reloads.

Table Linking: For texts classified as "table-answerable," a dedicated workflow allows annotators to link them to one or more tables from the main dataset. A modal-based preview system loads table details on demand to maintain performance.

A.3 Reasoning Step Annotation

Granular Annotation: For each question, annotators can define a multi-step reasoning process. Each step includes an operation (e.g., retrieve, sum), an explanation, and a list of operands.

Interactive Operand Selection: A dedicated interface allows annotators to build the operands list by:

- Clicking on cells directly within the source table.
- Selecting the result of a previous reasoning step.

- Entering a constant value.

Data Integrity: The system enforces operand limits based on the selected operation and provides clear visual feedback on the annotation detail page, where clicking on reasoning steps highlights the corresponding cells in the source table.

A.4 Automation and Auditing

Prompt Generation: The tool includes multiple prompt generators that read from .txt templates and dynamically insert table data, context paragraphs, and existing questions to create tailored prompts for LLMs.

Automated QA Ingestion: Annotators can use a "Process QA File" feature to automatically find a txt file, parse its JSON content, and add the new questions to the corresponding table and reference text entries, deleting the file upon success.

Intelligent JSON Parsing: The file processing logic is equipped with a robust parser that can automatically fix common LLM errors, such as trailing commas, unclosed quotes, and missing closing brackets/braces.

Data Audit Workflow: A comprehensive audit script can be run from the UI. It validates every table-sourced operand in the dataset by checking its fact against the value at its specified location. It can diagnose various off-by-one and formatting errors, assigning a validity status to each question and operand.

Audit Visualization: The results of the audit are displayed in a dedicated statistics menu, featuring a bar chart that visualizes the distribution of annotation validity statuses, allowing for targeted review and correction.

Part 3

Appendices

Code Repository

Code Repository

The code for this research is documented in a publicly available GitHub repository <https://github.com/mihoku/IndoMPF> which contains the following codes:

1. Dataset preparation process ([00_IndoMPF_Dataset](#))
This repository stores the data preparation workpapers and source code. i.e.:
 - a) Dataset extraction ([001_Dataset_Extraction](#))
 - b) Annotation platform web application ([002_Annotation-Platform-App](#))
 - c) Reference text classification model training ([003_Reference-Text-Categorization-BERT](#))
 - d) Initial table question-answering annotation generated using LLM ([004_Question-Answering-Annotation](#))
 - e) The final IndoMPF dataset after evaluation and correction ([005_IndoMPF-Dataset](#))
2. IndoMPFGATe model and training code ([01_IndoMPFGATe](#))
3. Embedding model finetuning for table retrieval ([02_Table_RAG_Gemma_MNR_Finetuning](#))
4. LLM Benchmark for IndoMPF Dataset ([03_LLMBenchmark](#))
5. Text generation model instruction tuning ([04_LLM_Instruction_Tuning](#))

Details for Question-Answer Annotation Platform

Custom Annotation Tool for a Hybrid Question-Answering Dataset

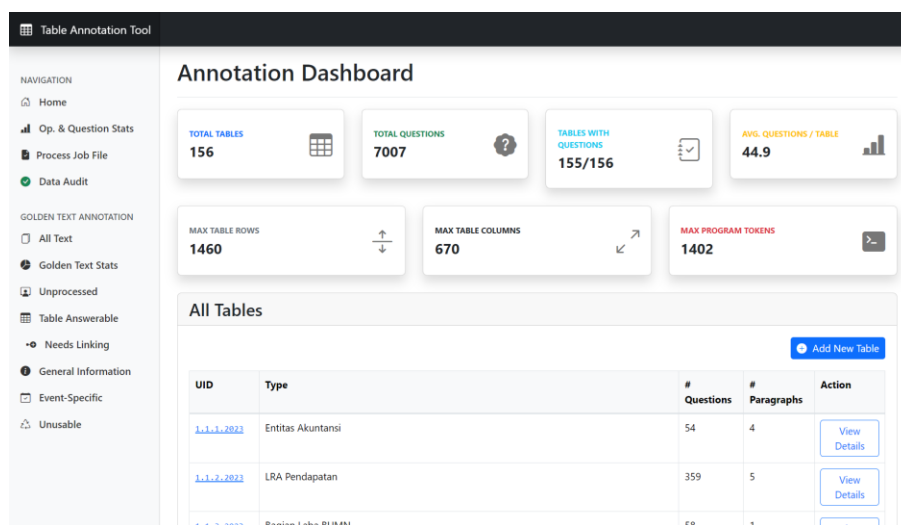
To support the research and development of a novel question-answering dataset that integrates both tabular and textual data, a custom, web-based annotation tool was developed. The primary objective of this tool is to provide a comprehensive and efficient environment for a multi-stage annotation process, which includes:

- Managing and viewing complex tabular data.
- Classifying and curating unstructured "golden text" sentences.
- Creating and linking questions to both tables and golden texts.
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A. Table and Question Management

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- **Table Detail View:** A dedicated page for each table that displays its metadata, a collapsible view of its content, associated paragraphs, and a list of all its questions.

Table Annotation Tool

Add Comment

NAVIGATION

Home

Op. & Question Stats

Process Job File

Data Audit

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Unprocessed

Table Answerable

Needs Linking

General Information

Event-Specific

Unusable

LRA Pendapatan

Table: LAPORAN REALISASI ANGGARAN PENDAPATAN PEMERINTAH PUSAT TAHUN ANGGARAN 2023 (dalam rupiah)

UID: 1.1.2.2023

Add New Question

Add Multiple Questions (JSON)

View Full JSON

Generate Prompt

Upload/Replace Table Data

Auto-Generate Questions

Process QA File

Delete All Questions

Show / Hide Table Data

Context Paragraphs

1. Pendapatan-LRA adalah semua penerimaan rekening KUN yang menambah Saldo Anggaran Lebih (SAL) dalam periode tahun anggaran yang bersangkutan, yang menjadi hak pemerintah dan tidak perlu dibayar kembali. Pendapatan diakui pada saat kas diterima pada Kas Umum Negara (KUN) atau pada saat terjadinya pengesahan atas pendapatan melalui dokumen pengesahan oleh KPPN selaku Kuasa BUN. Pendapatan yang diakui pada saat pengesahan oleh KPPN antara lain Pendapatan dari Hibah Langsung pada K/L dan Pendapatan BLU. Pendapatan-LRA disajikan dalam laporan realisasi anggaran entitas pelaporan/akuntansi.

2. Pendapatan LRA disajikan sesuai dengan jenis pendapatan yaitu Pendapatan Perpajakan-LRA, Pendapatan PNPB-LRA serta Pendapatan Hibah-LRA. Akuntansi Pendapatan-LRA dilaksanakan berdasarkan asas bruto, yaitu pendapatan dicatat tanpa dikurangkan/dikompensasikan dengan belanja yang dikeluarkan untuk memperoleh pendapatan tersebut. Pengeluaran asas bruto dapat terjadi jika penerimaan kas dari pendapatan tersebut lebih mencerminkan aktivitas pihak lain dari pada pemerintah atau penerimaan kas tersebut berasal dari transaksi yang perputarannya cepat, volume transaksi banyak dan jangka waktunya singkat.

3. Dalam LRA Pendapatan: Kode Akun mencerminkan pengelompokan jenis pendapatan: Pendapatan Negara (dengan kode akun 4) terdiri atas

Table Annotation Tool

Add Comment

NAVIGATION

Home

Op. & Question Stats

Process Job File

Data Audit

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Unprocessed

Table Answerable

Needs Linking

General Information

Event-Specific

Unusable

Linked Golden Texts

Link Golden Text

Hingga akhir tahun 2023, Pendapatan Negara mencapai Rp2.783,93 triliun, atau meningkat 5,62 persen (yoy) yang berasal dari Penerimaan Perpajakan, PNPB, dan Penerimaan Hibah yang melampaui target sedangkan realisasi Belanja Negara mencapai Rp3.121,22 triliun atau meningkat 0,81 persen (yoy).

UID: 174632ed-25c3-41d5-972e-7137f3e5e318

Details Unlink

Realisasi Pendapatan Negara hingga akhir tahun 2023 mencapai Rp2.783,93 triliun atau 105,56 persen dari target yang ditetapkan pada Perpres Nomor 75 tahun 2023, yang terdiri dari Pendapatan Perpajakan sebesar Rp2.154,21 triliun, PNPB sebesar Rp612,54 triliun, dan Pendapatan Hibah sebesar Rp17,18 triliun.

UID: d9e42097-8d70-4c77-b48a-02d97f09ca02

Details Unlink

Berdasarkan pertumbuhannya, realisasi Pendapatan Perpajakan dan PNPB tumbuh berturut-turut sebesar 5,88 persen (yoy) dan 2,84 persen (yoy) serta Pendapatan Hibah yang meningkat signifikan sebesar 201,69 persen (yoy).

UID: 4328ada1-d95b-40b7-bb0d-1075083e13c6

Details Unlink

Sedangkan, kinerja PNPB Tahun 2023 dipengaruhi kenaikan tarif royalti batubara dan kenaikan setoran yang berasal dari dividen Badan Usaha Milik Negara (BUMN), baik sektor perbankan maupun sektor non-Perbankan dan kontraksi Pendapatan SDA akibat moderasi harga komoditas terutama minyak bumi dan batubara.

UID: f9e080ee-20fd-463a-aa25-50cbf640db12

Details Unlink

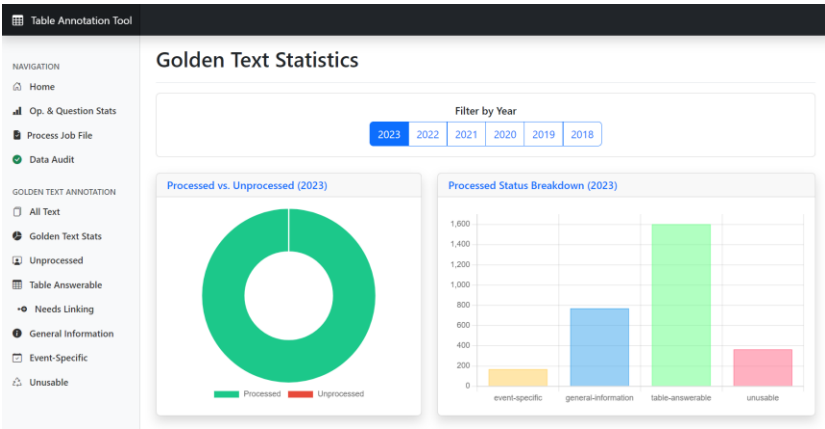
Table Annotation Tool			Add Comment
			Details Unlink
NAVIGATION Home Op. & Question Stats Process Job File Data Audit GOLDEN TEXT ANNOTATION All Text Golden Text Stats Unprocessed Table Answerable Needs Linking General Information Event-Specific Unusable			
Questions			329
UID	Question & Answer	Actions	
f65e44a70d19aeca5e9205da00d3955 matched facts (program parseable)	Berapa realisasi PENDAPATAN NEGARA pada tahun 2023? Realisasi PENDAPATAN NEGARA pada tahun 2023 adalah 2.783.929.676.930.200. 2.783.929.676.930.200	Edit	Delete
a025b2da74924432868087c0f4d7a961 matched facts (program parseable)	Apa uraian akun untuk Kode Akun 4112? Uraian akun untuk Kode Akun 4112 adalah Pendapatan Pajak Pertambahan Nilai. Pendapatan Pajak Pertambahan Nilai	Edit	Delete
724fbc667ca54c82a7b2e947a54eb80c matched facts (program parseable)	Berapa persen kenaikan/penurunan untuk PENDAPATAN PERPAJAKAN dari tahun 2022 ke 2023? Persentase kenaikan/penurunan untuk PENDAPATAN PERPAJAKAN dari tahun 2022 ke 2023 adalah 5,88%. 5,88%	Edit	Delete
1a04a67b555148b09dcafbdd3f6813dc matched facts (program parseable)	Berapa nilai APBN untuk Pendapatan Cukai Hasil Tembakau? Nilai APBN untuk Pendapatan Cukai Hasil Tembakau adalah 218.698.019.607.000. 218.698.019.607.000	Edit	Delete
2647fcb46f6941cc861c5809606d9517 matched facts (program parseable)	Apa kode akun untuk PENDAPATAN PAJAK PERDAGANGAN INTERNASIONAL? Kode akun untuk PENDAPATAN PAJAK PERDAGANGAN INTERNASIONAL adalah 412. 412	Edit	Delete

- **CRUD Operations:** The tool supports full Create, Read, Update, and Delete (CRUD) functionality for tables and questions through various menus, including form-based and JSON-based inputs.

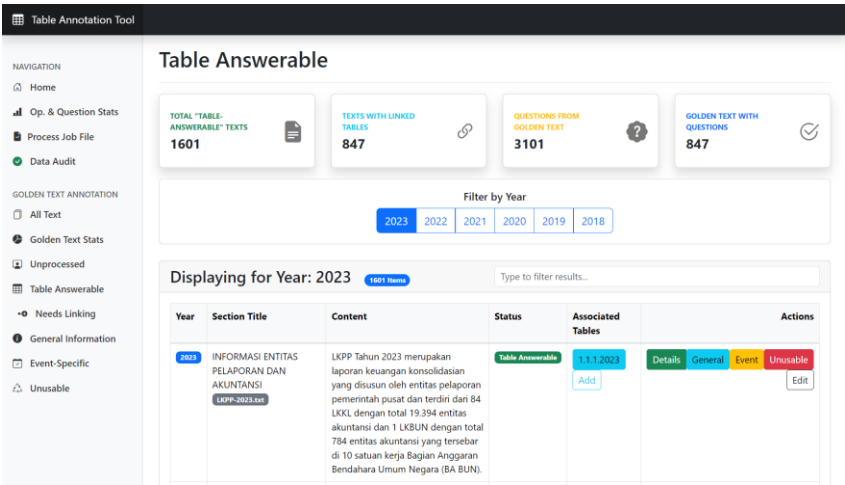
Table Annotation Tool	Add New Question
NAVIGATION Home Op. & Question Stats Process Job File Data Audit GOLDEN TEXT ANNOTATION All Text Golden Text Stats Unprocessed Table Answerable Needs Linking General Information Event-Specific Unusable	← Back to Table Detail Adding New Question for Table: 1.1.2.2023 Question Answer Sentence Answer(s) For multiple answers, enter each answer on a new line. Answer Type: span Scale: (No Scale)
Table Annotation Tool	Add Multiple Questions to Table 1.1.2.2023
NAVIGATION Home Op. & Question Stats Process Job File Data Audit GOLDEN TEXT ANNOTATION All Text Golden Text Stats Unprocessed Table Answerable Needs Linking General Information Event-Specific Unusable	Paste JSON List of Questions <pre>[{"-": "-"}]</pre>

B. Reference Text Annotation Workflow

- **Classification Menus:** A set of pages under the "Golden Text" section allows annotators to view reference texts based on their status (Unprocessed, Table Answerable, General Information, etc.).



- **Dynamic Classification:** On the "Unprocessed" page, annotators can classify a text with a single click. The action is handled via AJAX, providing a smooth UI experience without page reloads.



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Table Annotation Tool

NAVIGATION

Home

Op. & Question Stats

Process Job File

Data Audit

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Unprocessed

Table Answerable

Needs Linking

General Information

Event-Specific

Unusable

Unlinked Table-Answerable Texts

TOTAL "TABLE-ANSWERABLE" TEXTS

14272

TEXTS WITH LINKED TABLES

847

QUESTIONS FROM GOLDEN TEXT

3101

GOLDEN TEXT WITH QUESTIONS

847

Filter by Year

2023

2022

2021

2020

2019

2018

Displaying for Year: 2023

154 Items

Type to filter results...

Year	Section Title	Content	Status	Associated Tables	Actions
2023	EKONOMI MAKRO LAPP-2023.doc	Di tengah ketidakpastian global, kinerja manufaktur internasional mengalami pemulihan yang menyebabkan penurunan aktivitas perdagangan global.	Table Answerable	Add	Details General Event Unusable Edit
2023	EKONOMI MAKRO LAPP-2023.doc	Aktivitas manufaktur global juga masih melemah.	Table Answerable	Add	Details General Event Unusable Edit

C. Reasoning Step Annotation

- Granular Annotation:** For each question, annotators can define a multi-step reasoning process. Each step includes an operation (e.g., retrieve, sum), an explanation, and a list of operands.

Table Annotation Tool

NAVIGATION

Home

Op. & Question Stats

Process Job File

Data Audit

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Unprocessed

Table Answerable

Needs Linking

General Information

Event-Specific

Unusable

Add New Reasoning Step (1/2)

Back to Annotation Detail

Step 1: Define Operation and Explanation

Operation Type

-- Choose an operation --

Explanation

Explain what this reasoning step does...

Save and Select Operands --

© 2025 - Table Annotation Tool

Table Annotation Tool

NAVIGATION

Home

Op. & Question Stats

Process Job File

Data Audit

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Unprocessed

Table Answerable

Needs Linking

General Information

Event-Specific

Unusable

Edit Reasoning Step (2/2)

Question:

Apakah realisasi PENDAPATAN SUMBER DAYA ALAM tahun 2023 melebihi target APBN?

Now Annotating Step #0

"Mengambil nilai Realisasi 2023 untuk PENDAPATAN SUMBER DAYA ALAM."

Select Operand from Previous Step

No previous steps to select from.

Select Operand from Source Table (Click any cell)

Type to filter table...

Kode Akun	Uraian Akun	APBN	Realisasi 2023	% thd APBN
4	PENDAPATAN NEGARA	2.637.248.881.485.000	2.783.929.676.930.200	105,56%
41	PENDAPATAN PERPAJAKAN	2.118.348.000.000.000	2.154.208.219.467.830	101,69%
411	PENDAPATAN PAJAK	2.045.450.000.000.000	2.089.718.285.003.420	102,16%

Manage Operands for #0

Operation: add

Current Operands:

254.203.031.073.614 (table)

Add Constant as Operand

Enter constant...

Operand limit reached for this operation.

Finish Editing Step

- **Interactive Operand Selection:** A dedicated interface allows annotators to build the operands list by:
 - Clicking on cells directly within the source table.
 - Selecting the result of a previous reasoning step.
 - Entering a constant value.
- **Data Integrity:** The system enforces operand limits based on the selected operation and provides clear visual feedback on the annotation detail page, where clicking on reasoning steps highlights the corresponding cells in the source table.

The screenshot displays the 'Table Annotation Tool' interface. On the left is a navigation sidebar with options: Home, Op. & Question Stats, Process Job File, Data Audit (selected), Golden Text Annotation (All Text, Golden Text Stats, Unprocessed, Table Answerable, Needs Linking), General Information, Event-Specific, and Unusable. The main area shows a 'Validity Status: matched facts (program parseable)' with buttons for 'Back to Table Detail' and 'Edit JSON'. Below this is the 'Reasoning Steps' section, which includes a table with three columns: '# & Explanation', 'Operator', and 'Operands (Clickable)'. The table contains three steps: Step 0 (retrieving 2023 revenue), Step 1 (retrieving target APBN), and Step 2 (comparing revenue to target). Each step has a 'Result' and an 'Edit' button. At the bottom, the 'Source Table: LRA Pendapatan' is shown with a filter input.

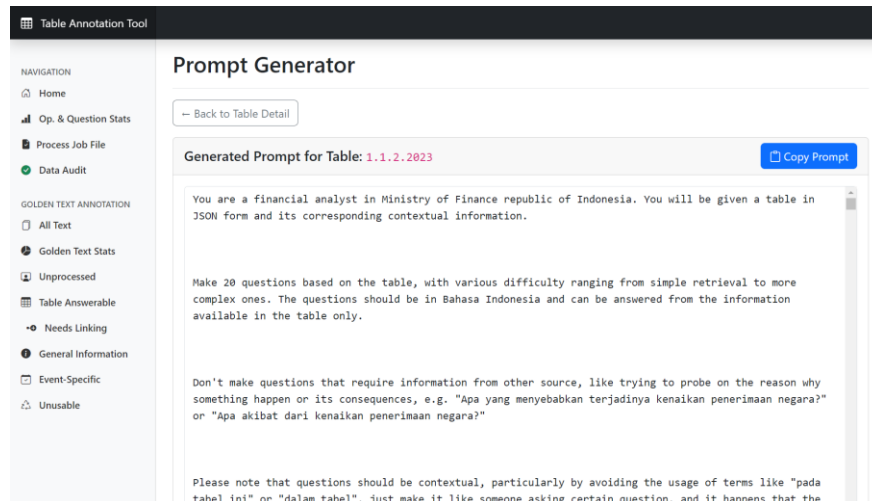
# & Explanation	Operator	Operands (Clickable)
#0: Mengambil nilai Realisasi 2023 untuk PENDAPATAN SUMBER DAYA ALAM. Result: 25403031073614	retrieve	254.203.031.073.614 [Edit]
#1: Mengambil nilai target APBN untuk PENDAPATAN SUMBER DAYA ALAM. Result: 223312141813000	retrieve	223.312.141.813.000 [Edit]
#2: Membandingkan apakah nilai realisasi lebih besar dari target APBN. Result: true	greater	254203031073614 223312141813000 [Edit]

Source Table: LRA Pendapatan

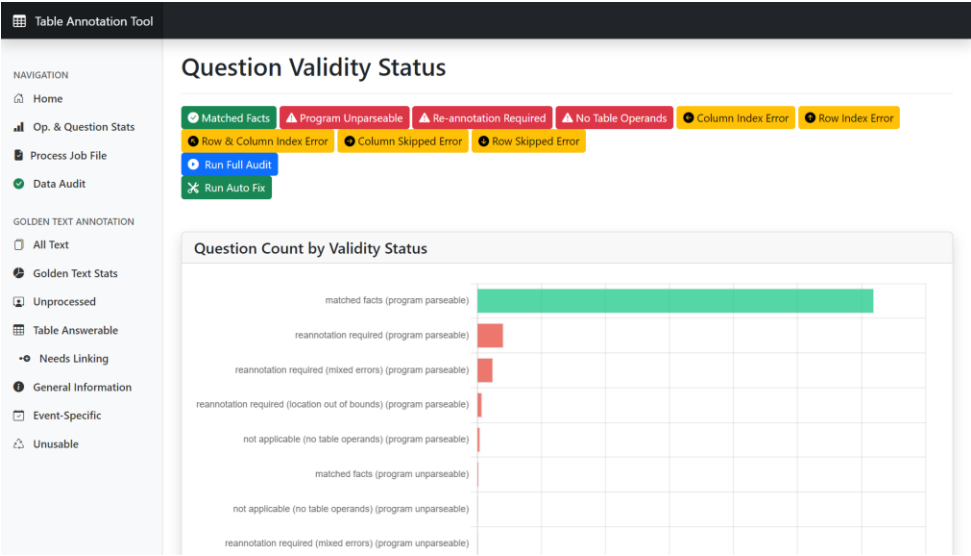
Type to filter table...

D. Automation and Auditing

- **Prompt Generation:** The tool includes multiple prompt generators that read from .txt templates and dynamically insert table data, context paragraphs, and existing questions to create tailored prompts for LLMs.



- **Automated QA Ingestion:** Annotators can use a "Process QA File" feature to automatically find a txt file, parse its JSON content, and add the new questions to the corresponding table and reference text entries, deleting the file upon success.
- **Intelligent JSON Parsing:** The file processing logic is equipped with a robust parser that can automatically fix common LLM errors, such as trailing commas, unclosed quotes, and missing closing brackets/braces.
- **Data Audit Workflow:** A comprehensive audit script can be run from the UI. It validates every table-sourced operand in the dataset by checking its fact against the value at its specified location. It can diagnose various off-by-one and formatting errors, assigning a validity status to each question and operand.
- **Audit Visualization:** The results of the audit are displayed in a dedicated statistics menu, featuring a bar chart that visualizes the distribution of annotation validity statuses, allowing for targeted review and correction.



Guidelines for Professional Evaluator

Guidelines for Professional Evaluators: IndoMPF Dataset

Project Title	Generating Notes to Government Financial Statements using Multi-step Reasoning and Tabular Graph Enhanced Language Model
Institution	Monash University
Research Context	Master Thesis by Reza Darmawan (Student ID: 33462135)

1. Purpose of Evaluation

Welcome to the evaluation team for the IndoMPF dataset. Your role is critical to our research. The primary goal of this project is to develop a high-quality dataset for training advanced AI models capable of complex reasoning over financial and tabular data.

Your mission is to **ensure the accuracy, logical consistency, and efficiency of the annotated reasoning steps** for each question in the dataset. A high-quality reasoning program is one that is not only correct but also follows the most direct and logical path to the answer. Your feedback will be instrumental in refining the dataset to a research-grade standard.

2. Your Core Task

Your main responsibility is to review the questions associated with each table and meticulously analyze their reasoning programs. You will identify and report any of the following issues:

- **Inaccuracy:** The reasoning program produces the wrong answer or uses incorrect data points (operands).
- **Inefficiency:** The program arrives at the correct answer but uses a convoluted or unnecessarily complex series of steps. For example, using three addition steps when a single sum would suffice.
- **Logical Flaws:** A step in the reasoning does not logically follow from the previous ones.
- **Ambiguity:** The explanation for a reasoning step is unclear or does not accurately describe the operation being performed.

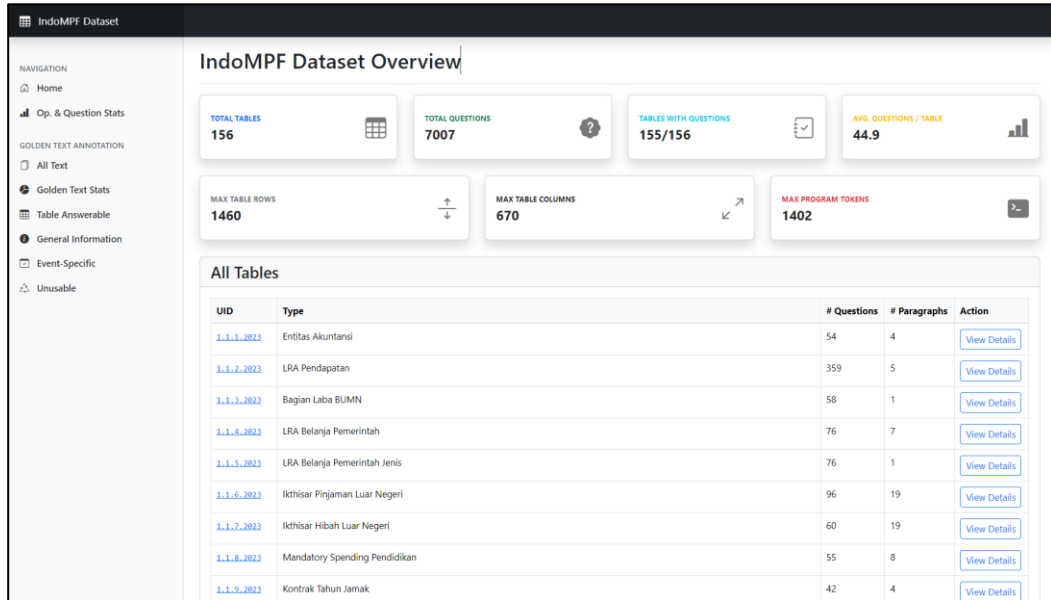
3. Step-by-Step Evaluation Workflow

Please follow this workflow to navigate the tool and perform your evaluation.

a) **Login.** You can access the web evaluation platform through the following link: (<https://rezadr.pythonanywhere.com/>). When you are asked to sign in, use this credential:

- **Username: evaluator**
- **Password: indompf**

b) **Start at the Dashboard:** From the main [Home](#) page, you will see a list of all tables in the dataset.



IndoMPF Dataset Overview

TOTAL TABLES 156

TOTAL QUESTIONS 7007

TABLES WITH QUESTIONS 155/156

AVG. QUESTIONS / TABLE 44.9

MAX TABLE ROWS 1460

MAX TABLE COLUMNS 670

MAX PROGRAM TOKENS 1402

All Tables

UID	Type	# Questions	# Paragraphs	Action
1.1.1.2023	Entitas Akuntansi	54	4	View Details
1.1.2.2023	LRA Pendapatan	359	5	View Details
1.1.3.2023	Bagian Laba BUMN	58	1	View Details
1.1.4.2023	LRA Belanja Pemerintah	76	7	View Details
1.1.5.2023	LRA Belanja Pemerintah Jenis	76	1	View Details
1.1.6.2023	Ikhtisar Pinjaman Luar Negeri	96	19	View Details
1.1.7.2023	Ikhtisar Hibah Luar Negeri	60	19	View Details
1.1.8.2023	Mandatory Spending Pendidikan	55	8	View Details
1.1.9.2023	Kontrak Tahun Jarak	42	4	View Details

c) **Select a Table:** Click on the UID of a table you wish to evaluate. This will take you to the **Table Detail** page.

IndoMPF Dataset

NAVIGATION

Home

Op. & Question Stats

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Table Answerable

General Information

Event-Specific

Unusable

IKHTISAR PINJAMAN LUAR NEGERI

Table: IKHTISAR LAPORAN KEUANGAN PINJAMAN LUAR NEGERI PER 31 DESEMBER 2023

View Full JSON

Show / Hide Table Data

Context Paragraphs

- Kolom 'BA' berisi Bagian Anggaran yang bertanggung jawab atas pelaksanaan proyek yang dibiayai pinjaman luar negeri tersebut.
- Kolom 'Pemberi Pinjaman' adalah entitas atau badan dari luar negeri yang memberikan pinjaman kepada Pemerintah Indonesia.
- Kolom 'Proyek' berisi nama kegiatan atau program yang pelaksanaannya didanai oleh pinjaman luar negeri.
- Kolom 'Loan_ID' berisi nomor identifikasi pinjaman luar negeri di sistem pengelolaan pinjaman pemerintah Indonesia.
- Kolom 'Register' berisi nomor yang diberikan untuk mengidentifikasi pinjaman luar negeri dalam sistem administrasi pinjaman/hibah.
- Kolom 'Date Signed' berisi tanggal penandatanganan perjanjian pinjaman luar negeri (loan agreement).
- Kolom 'Date_Effective' berisi tanggal perjanjian pinjaman luar negeri sudah dapat dilaksanakan, termasuk di dalamnya realisasi (execution) proyek dan pencairan (disbursement) pinjaman.
- Kolom 'Closing Date' adalah batas akhir waktu untuk pencairan dan/atau penarikan dana Pinjaman Luar Negeri melalui penerbitan Surat Perintah Pencairan Dana dan/atau surat pengantar - Surat Penarikan Dana oleh KPKN atau batas akhir waktu transaksi pembayaran dari Pemberi Pinjaman Luar Negeri kepada Pemerintah.
- Kolom 'Loan Amount' adalah nilai pinjaman luar negeri sesuai dengan perjanjian pinjaman (loan agreement). Di dalam kolom ini juga terdapat mata uang denominasi pinjaman luar negeri tersebut.
- Kolom 'Neraca (Rupiah)' berisi aset dan kewajiban yang berkaitan dengan pelaksanaan proyek yang dibiayai oleh pinjaman luar negeri. Terdiri atas 'Aset Lancar' yang dapat berupa kas dan setara kas atau aset yang memiliki masa manfaat atau akan habis digunakan dalam jangka waktu 12 bulan, 'Aset Tetap' yaitu aset berwujud yang memiliki masa manfaat lebih dari 12 bulan, dan 'Aset Lainnya' yaitu aset tidak berwujud dengan masa manfaat lebih dari 12 bulan.
- Kolom 'Kementerian/Lembaga selaku Executing Agency' adalah Kementerian Negara/Lembaga selaku unit teknis yang menjadi koordinator atas pelaksanaan kegiatan yang dibiayai dari Pinjaman Luar Negeri, dan bertanggung jawab sesuai tugas dan fungsinya.
- Kolom 'Pemeriksa (BPK/BPKP/KAP)' menginformasikan pihak yang melakukan audit atas proyek yang dibiayai melalui pinjaman luar negeri tersebut.
- Kolom 'Disbursement s.d. 2022' berisi total pencairan pinjaman luar negeri sampai dengan tahun 2022. Karena memiliki denominasi yang berbeda-beda, maka kolom 'Disbursement s.d. 2022' pada tabel tidak dapat dijumlahkan secara total.

d) **Review the Table and Questions:** On the Table Detail page, familiarize yourself with the table's content and review the list of questions at the bottom.

IndoMPF Dataset

NAVIGATION

Home

Op. & Question Stats

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Table Answerable

General Information

Event-Specific

Unusable

Questions

UID	Question & Answer
846744803aff410b8ba29725133805f matched facts (program parsable)	Siapaakah pemberi pinjaman untuk proyek "PATIMBAN PORT DEV. PROJECT (I)"? Pemberi pinjaman untuk proyek "PATIMBAN PORT DEV. PROJECT (I)" adalah JICA. JICA
2c8792c3ba338edaf1c9f4ac23d35e72 matched facts (program parsable)	Berapa total nilai pinjaman (LOAN AMOUNT) dalam mata uang JPY untuk proyek "E/S JKT MRT EAST WEST PHASE I"? Total nilai pinjaman dalam mata uang JPY untuk proyek "E/S JKT MRT EAST WEST PHASE I" adalah 1.919.000.000. 1.919.000.000
c13d72a0b7f6d7789ba38bc3aaf34fcad matched facts (program parsable)	Kapan tanggal efektif (DATE EFFECTIVE) untuk pinjaman proyek "BIG DATA INTELLIGENCE"? Tanggal efektif untuk pinjaman proyek "BIG DATA INTELLIGENCE" adalah 13/11/2023. 13/11/2023
87c63820612a4a218c8a7dc46b0462c5 matched facts (program parsable)	Sebutkan lembaga yang menjadi Executing Agency untuk proyek NUDP (No. 114). Lembaga yang menjadi Executing Agency untuk proyek NUDP (No. 114) adalah Kementerian PUPR. Kementerian PUPR
2937886ff8f640778d4c83b599de4aad matched facts (program parsable)	Berapa nilai Belanja Modal dalam Rupiah pada tahun 2023 untuk proyek "TRMV BARESKIM POLRI" (No. 153)? Nilai Belanja Modal pada tahun 2023 untuk proyek "TRMV BARESKIM POLRI" adalah 2.101.642.265.618 Rupiah. 2.101.642.265.618
fc7d2aab4d70413182637ccfc08f906 matched facts (program parsable)	Sebutkan nama proyek yang didanai oleh EXIM BANK OF CHINA dengan LOAN_ID 28761000. Proyek yang didanai oleh EXIM BANK OF CHINA dengan LOAN_ID 28761000 adalah Toll Road Serang-Panimbang. Toll Road Serang-Panimbang
961a634c15cc0ba39b5798202b95633 matched facts (program parsable)	Apa mata uang pinjaman untuk proyek "Hasanuddin University Hospital" (No. 71)? Mata uang pinjaman untuk proyek "Hasanuddin University Hospital" adalah EUR. EUR
7c94bf03f0b41f7ba3a0d5413a31081 matched facts (program parsable)	Siapaakah pemeriksa (auditor) untuk proyek "Trans South-South Java Road" (No. 112)? Pemeriksa untuk proyek "Trans South-South Java Road" adalah BPKP. BPKP

e) **Analyze a Question:** Click on a question's text to navigate to the **Annotation Detail** page. This is where you will perform the core of your analysis.

3

IndoMPF Dataset

NAVIGATION

Home

Op. & Question Stats

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Table Answerable

General Information

Event-Specific

Unusable

Question Details

Hitung total pencairan pinjaman pada tahun 2023 (DISBURSEMENT 2023) untuk semua proyek yang didanai oleh IBRD dalam mata uang USD.

UID: 8da01cc74c6f44369417e9b9811b1a28

Final Answer: 535.151.723

Answer Type: mathematic

Scale: N/A

Validity Status: matched facts (program parsable)

Back to Table Detail

Reasoning Steps

# & Explanation	Operator	Operands (Clickable)
#0: Menjumlahkan semua 'DISBURSEMENT 2023' untuk proyek yang didanai IBRD dengan mata uang USD.	sum	500.000 28.000.000 82.000.000 83.930 13.000.000 116.750.000 1.810.200 355.296 116.750.000 1.237.000 23.499.500 88.000.000 500.000 71.900.000 2.815.000 28.000.000 13.000.000 104.942.575 116.750.000 2.152.722 20.120.000 64.500.000

| Result: 535.151.723 | | |

Source Table: Ikhtisar Pinjaman Luar Negeri

Type to filter table...

4. How to Analyze a Reasoning Program

The **Annotation Detail** page is highly interactive to help you trace the logic of a question's reasoning.

- To Understand the Big Picture:** Click anywhere on the **row of a reasoning step** (e.g., the row for #1). This will highlight **all** the cells in the source table that were used as operands in explanation that specific step. This is useful for quickly seeing the data context.

IndoMPF Dataset

NAVIGATION

Home

Op. & Question Stats

GOLDEN TEXT ANNOTATION

All Text

Golden Text Stats

Table Answerable

General Information

Event-Specific

Unusable

Validity Status: matched facts (program parsable)

Back to Table Detail

Reasoning Steps

# & Explanation	Operator	Operands (Clickable)
#0: Mengambil tanggal efektif dari baris yang sesuai dengan nama proyek 'BIG DATA INTELLIGENCE'.	extracted	13/11/2023

| Result: 13/11/2023 | | |

Source Table: Ikhtisar Pinjaman Luar Negeri

Type to filter table...

No	BA	PHLN	PEMBERI PINJAMAN	PROYEK	LOAN_ID	REGISTER	DATE_SIGNED	DATE_EFFECTIVE	CLOSING DATE	LOAN_AMOUNT	LC
1	006	LK	BNI New York PLN 1	BIG DATA INTELLIGENCE	24097050	11U4M6GA	02/03/2023	13/11/2023	02/09/2024	USD	84
2	006	LK	BNI New York PLN 1	BIG DATA INTELLIGENCE	24097050	11U4M6GA	02/03/2023	13/11/2023	02/09/2024	USD	84

4

- b) **To Verify a Specific Operand:** Click on an individual **operand button** within a reasoning step.
- If the operand is from the table (a green button), it will highlight **only that single cell** in the source table.

The screenshot shows the IndoMPF Dataset interface. The 'Reasoning Steps' section displays a step with the explanation: '#0: Mengambil tanggal efektif dari baris yang sesuai dengan nama proyek 'BIG DATA INTELLIGENCE''. The operator is 'return' and the operand is '13/11/2023'. Below this, the 'Source Table: Ikhtisar Pinjaman Luar Negeri' is shown. The table has columns: No, BA, PHLN, PEMBERI PINJAMAN, PROYEK, LOAN_ID, REGISTER, DATE_SIGNED, DATE_EFFECTIVE, CLOSING DATE, LOAN_AMOUNT, and LC. The row for 'BIG DATA INTELLIGENCE' is highlighted, with the 'DATE_EFFECTIVE' cell (13/11/2023) specifically highlighted in yellow.

- If the operand is a reference to a previous step (a yellow button, e.g., #0), it will highlight the **entire row for step #0** in the reasoning table above.

The screenshot shows the IndoMPF Dataset interface. The 'Reasoning Steps' section displays a step with the explanation: '#0: Menjumlahkan total pencairan hingga 2022 dan pencairan pada 2023 untuk proyek PROC. OF FRIGATE VESSEL'. The operator is 'sum' and the operand is '#0'. Below this, the 'Source Table: Ikhtisar Pinjaman Luar Negeri' is shown. The table has columns: No, BA, PHLN, PEMBERI PINJAMAN, PROYEK, LOAN_ID, REGISTER, DATE_SIGNED, DATE_EFFECTIVE, CLOSING DATE, LOAN_AMOUNT, LOAN_AMOUNT, and DISB S.D.2. The row for 'PROC. OF FRIGATE VESSEL' is highlighted, with the 'LOAN_AMOUNT' cell (612,000,000) specifically highlighted in yellow.

Use these interactive features to trace the flow of logic from one step to the next and to verify that each operand is referencing the correct data.

5. How to Report Findings: The Comment System

To report your findings, you will use the built-in asynchronous commenting system.

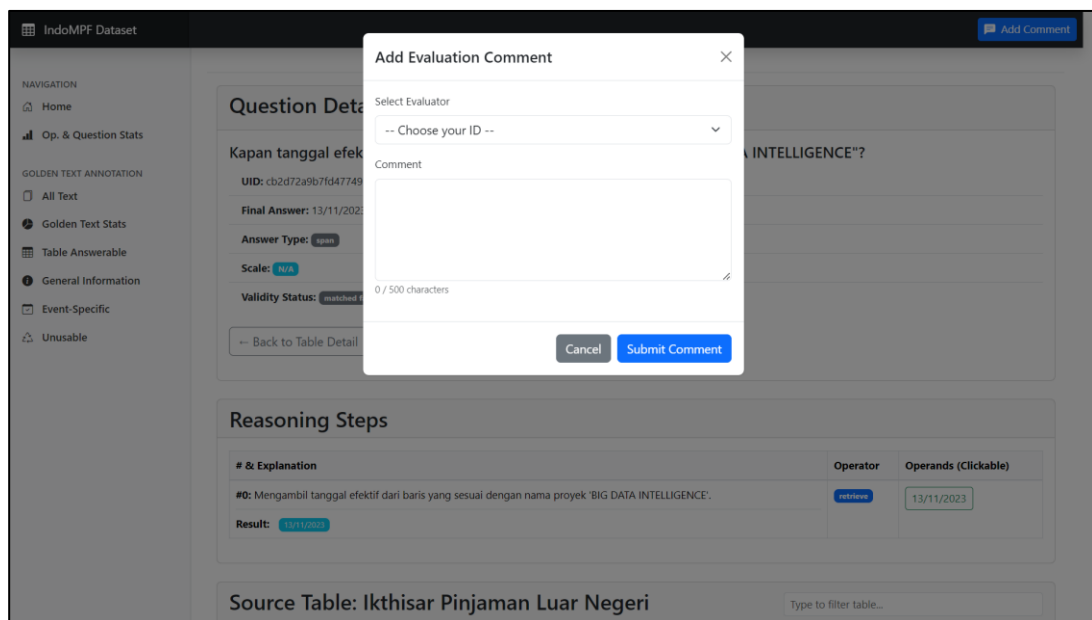
The "Add Comment" Button



On any **Table Detail** or **Question Annotation Detail** page, you will see a blue "Add Comment" button in the top navigation bar. This button will follow you as you scroll.

Submitting a Comment

- Click the **"Add Comment"** button to open a modal window.
- Select Your ID:** Choose your assigned ID (Evaluator 1, Evaluator 2, or Evaluator 3) from the dropdown menu. This is mandatory for every comment.



- Write Your Comment:** Provide a clear and concise description of your finding in the text area (max 500 characters). Be specific about the issue you found.
- Submit:** Click "Submit Comment". The modal will close, and your comment will be saved instantly without reloading the page.

Automatic Context Tagging

The system automatically knows where you are.

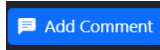
- If you submit a comment from a **Table Detail** page, it will be tagged with the `table_uid`.
- If you submit from a **Question Annotation Detail** page, it will be tagged with both the `table_uid` and the `question_uid`.

6. Marking a Table as Reviewed

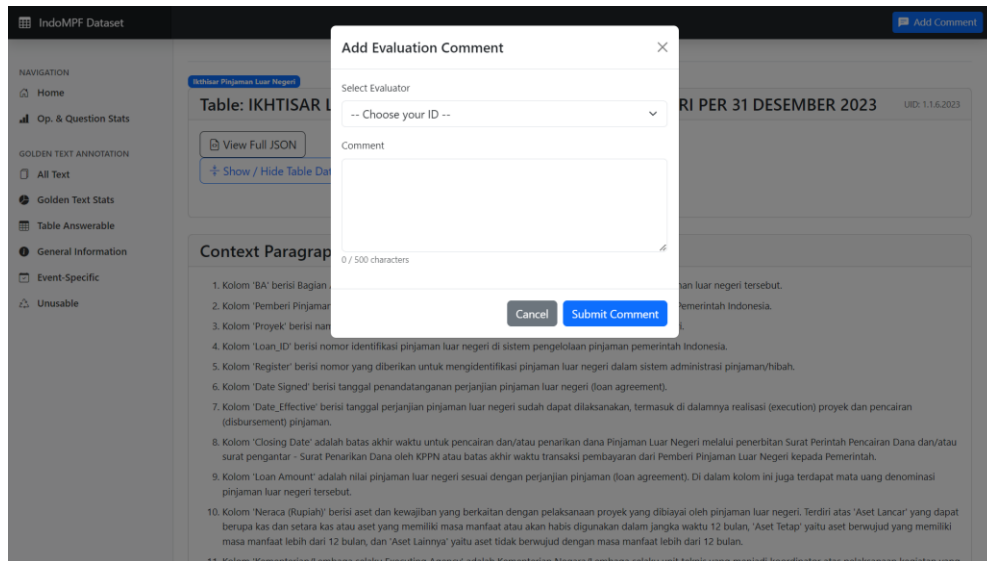
Once you have thoroughly reviewed **all** the questions for a particular table, please perform one final action:

a) Navigate back to the **Table Detail** page for that table.

b) Open the **"Add Comment"** modal.



c) Select your Evaluator ID.



d) In the comment box, please write: **"Review for this table complete."**

This final comment serves as your sign-off, indicating that the table and all its associated questions have been fully evaluated by you.

Thank you for your expertise and diligence. Your contribution is invaluable to this research.